Online Appendix

Night and Day: Diurnal Warming and Structural Transformation in India

Vedarshi Shastry

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This appendix has four parts. Appendix A develops the general-equilibrium framework behind the diurnal decomposition, with proofs, and closes with a single table mapping each prediction to the primitive that carries it and to the estimate it is checked against. Appendix B derives the panel estimator and the two standard-error families, selects the inference procedure by its simulated size, and collects the diagnostics on the diurnal basis itself: collinearity of the temperature pair, the mean-and-range reparameterization, and the Conley radius sweep. Appendix C documents the construction of each sample and reports the coefficients behind the main-text figures. Appendix D collects the robustness checks the main text cites. Throughout, the main-text baseline and the tables behind its figures use surface-air minima and maxima (T_{min}, T_{max}); the generated robustness tables of Appendix D default to the wet-bulb counterparts (T_{wmin}, T_{wmax}) unless a table notes otherwise, so that the two measures can be read against one another.

The organization of this online appendix has significantly benefited from AI/agentic research assistance in systematic bookkeeping of exhibits, formalising the model and proofs in TeX, and producing a meta-analysis of the empirical approach. All errors remain my own.

Appendix A. Theoretical Framework

The empirical decomposition rests on a two-sector general-equilibrium framework that assigns nighttime warming ($T_{\min}$) to a land-biased shock and daytime warming ($T_{\max}$) to a Hicks-neutral productivity shock, embedded in a constant-elasticity-of-substitution (CES) agricultural technology under two market closures. This appendix develops the model, derives all comparative statics from first principles, and delimits what the framework can and cannot deliver.

A.1. Overview and Relation to the Empirics

The empirical decomposition isolates two sets of facts that any theoretical framework must confront (Table A1 at the end of this section summarizes them against the model's predictions). On the agricultural side, nighttime and daytime warming contract cultivated area by indistinguishable amounts, whereas their effects on crop yields and harvest prices diverge: in the decadal panel, only daytime warming raises the local harvest price ($F = 13.34$), while over the two-decade long difference, daytime warming depresses cereal yields relative to nighttime warming ($F = 4.56$), and under humidity-adjusted (wet-bulb) temperatures nighttime warming raises yields outright with area intact—the signature of land augmentation. On the labor side, the two margins carry *opposite* signs: a warmer night raises the agricultural wage-labor share and drains the cultivator and seasonal pools, whereas a warmer day lowers it and refills the seasonal margin. Consequently, when reparameterized into the diurnal mean and diurnal temperature range (DTR), the common land contraction loads entirely onto the mean while the opposite-signed labor reallocation loads entirely onto the range. The labor divergence is sharp in rural district populations and muted in urban ones, and in Census towns *both* margins depress non-agricultural employment.

Three modeling choices are dictated by these facts:

1. **CES production with gross substitutes ($\sigma > 1$), not Cobb–Douglas.** A land-biased shock is a distinct economic object from a neutral productivity shock only if labor and land substitute with an elasticity $\sigma \neq 1$. Under Cobb–Douglas ($\sigma = 1$), a land-augmenting shock is observationally equivalent to a rescaled neutral shock (Lemma A2), leaving no diurnal divergence in relative labor responses. We maintain $\sigma > 1$ (gross substitutes), the empirically relevant case for labor–land substitution through cultivation intensity.
2. **Land-biased $T_{\min}$ shocks versus neutral (or labor-inclusive) $T_{\max}$ shocks.** Because crops stand in the field 24 hours a day while field workers work predominantly during daylight hours, nighttime warming acts on the land factor—altering the productive services of soil and crops on cultivated parcels and shifting the extensive threshold at which marginal land breaks even. Daytime heat reduces both crop and outdoor worker productivity, acting as a Hicks-neutral (or labor-disaugmenting) shock across the farm operation (Section A.3).
3. **Two food-market closures, one technology.** Rural and urban district aggregates share the same agricultural production technology but potentially differ in food-market integration. We nest a closed subsistence economy (Stone–Geary preferences with local price clearing) and an open price-taking economy within a single openness parameter $\chi \in [0, 1]$.

A central caution. A tempting reading of the rural/urban contrast is that rural districts behave as closed economies and urban areas as open ones, and that this difference in market closure *generates* the divergence in signs. Sections A.5–A.6 prove otherwise: the *sign* of the labor divergence is a factor-bias property of the open (or partially open) economy, whereas a closed subsistence economy *reverses* both signs (Proposition A4). Because nighttime warming raises agricultural wage labor in

rural and urban populations alike—the binding condition for openness by Proposition A5—the data select the open regime ($\chi < \chi^*$) in both populations. The urban daytime estimate, a statistical zero ($+0.006$, $t = 0.32$), sits within sampling error of the common day/night ratio implied by the open economy (≈ -0.024). Market closure governs the harvest-price response (when $0 < \chi < \chi^*$), while the rural/urban difference in labor response magnitudes is governed by sectoral scale and seasonal-pool depth (Section A.9).

A.2. Environment

A local economy (a Census district, or its rural, urban, or town sub-economy) produces an agricultural good a and a non-agricultural good n. The non-agricultural good serves as the numeraire ($p_n \equiv 1$), and $p \equiv p_a$ denotes the relative price of the agricultural good. Throughout, a hat over any variable denotes a log deviation from an initial equilibrium, $\hat{x} \equiv d \ln x$, and all comparative statics are derived to first order.

A.2.1. Agricultural technology

Agricultural output Y_a combines agricultural field labor L_a and effective land BZ through a CES production function scaled by total factor productivity A:

$$Y_a \;=\; A\Big[\,\theta\, L_a^{\rho} + (1-\theta)\big(BZ\big)^{\rho}\,\Big]^{1/\rho}, \qquad \rho \;\equiv\; \frac{\sigma-1}{\sigma}, \quad \theta \in (0,1). \tag{A1}$$

Here $\sigma > 0$ is the elasticity of substitution between labor and land, Z is physical cultivated area (an extensive margin chosen by the farm; Section A.3.2), and BZ is *effective land*, where B scales the productive services of a unit of cultivated area. The two diurnal temperature margins enter through distinct technological channels:

$$A = A(T_{\max}), \qquad B = B(T_{\min}), \tag{A2}$$

where $A(T_{\max})$ is a Hicks-neutral total factor productivity shifter and $B(T_{\min})$ is a land-biased technology index.

ASSUMPTION A1 (Technology and diurnal shocks).

(i) Gross substitutes: $\sigma > 1$ *(so* $\rho \in (0,1)$*).*

(ii) Daytime heat lowers TFP: $A'(T_{\max}) < 0$*.*

(iii) Nighttime warming is land-biased: $B'(T_{\min}) \gtrless 0$*. As our benchmark we take* $B'(T_{\min}) \geq 0$*, the sign corroborated by the significantly positive humidity-adjusted (wet-bulb) night yield and by the positive long-difference night yield relative to daytime warming (Remark A7).*

In log-deviation form ($\hat{x} \equiv d \ln x$), the exogenous climate shocks to neutral TFP and land-augmenting productivity are written directly with hats:

$$\hat{A} \;\equiv\; \frac{A'(T_{\max})}{A}\, dT_{\max} \;\leq\; 0, \qquad \hat{B} \;\equiv\; \frac{B'(T_{\min})}{B}\, dT_{\min} \;\gtrless\; 0 \quad (\geq 0 \text{ benchmark}). \tag{A3}$$

Let s_L and $s_Z = 1 - s_L$ denote the equilibrium cost (and revenue) shares of labor and effective

land at the initial equilibrium:

$$s_L \equiv \frac{\theta L_a^{\rho}}{\theta L_a^{\rho} + (1-\theta)(BZ)^{\rho}} \in (0,1), \qquad s_Z \equiv 1 - s_L. \tag{A4}$$

Log-differentiating (A1) yields the standard first-order output and crop-yield ($y_a \equiv Y_a/Z$) identities:

$$\hat{Y}_a = \hat{A} + s_L \hat{L}_a + s_Z\big(\hat{B} + \hat{Z}\big), \qquad \hat{y}_a \equiv \hat{Y}_a - \hat{Z} = \hat{A} + s_L\big(\hat{L}_a - \hat{Z}\big) + s_Z \hat{B}. \tag{A5}$$

Finally, define the real-wage elasticity of agricultural labor demand at fixed land as

$$\varepsilon_a \equiv \frac{\sigma}{s_Z} > 0. \tag{A6}$$

A.2.2. Intensive versus extensive roles of land

Nighttime warming acts on agricultural land along two conceptually distinct margins. The *intensive* role is the land-augmentation index $B(T_{\min})$ in (A1), which alters the productive services of land remaining in cultivation and enters crop yields $\hat{y}_a$. The *extensive* role operates on viable cultivable area: thermal and evaporative stress can push marginal parcels below the break-even threshold for cultivation. We model this in Section A.3.2 via a convex cultivation cost $C(Z)$ subject to climate-induced cost shifters $\psi_T\, dT \geq 0$ for $T \in \{T_{\min}, T_{\max}\}$.

Keeping these two roles distinct clarifies the difference between dry-bulb and wet-bulb temperature estimates. Under *wet-bulb* (humidity-adjusted) nighttime warming, the intensive augmentation channel is isolated: yields rise significantly and cultivated area does not contract (Remark A7). Under *dry-bulb* surface-air warming, the nighttime margin contracts cultivated area alongside daytime warming because dry-bulb $T_{\min}$ loads an evaporative moisture-stress component on marginal parcels that the wet-bulb transformation nets out; even in dry-bulb, however, the two-decade long difference reveals a significant yield divergence between nighttime and daytime warming ($\Delta T_{\min} : +0.052$ vs. $\Delta T_{\max} : -0.178$, $F = 4.56$).

A.2.3. Non-agricultural sector

The non-agricultural sector produces the numeraire with decreasing returns to labor:

$$Y_n = A_n\, L_n^{\nu}, \qquad \nu \in (0,1), \qquad A_n = A_n(T_{\min}, T_{\max}), \tag{A7}$$

where $\nu < 1$ reflects congestion on fixed local urban/non-farm capital and generates a downward-sloping non-agricultural labor demand schedule with wage elasticity $\varepsilon_n \equiv \frac{1}{1-\nu} > 1$. Non-agricultural productivity A_n is weakly decreasing in both temperature margins ($\hat{A}_n \leq 0$) due to the physiological toll of heat and sleep disruption on physical, outdoor informal work. Where agriculture dominates employment (rural district populations), the within-agriculture factor-bias reallocation dominates and $\hat{A}_n$ is second order, so we set $\hat{A}_n = 0$ as our rural benchmark (and verify in Remark A8 that allowing $\hat{A}_n < 0$ does not drive the rural sign test). Where agriculture is a minor sector (Census towns, Section A.10), $\hat{A}_n < 0$ becomes the primary observable channel.

A.2.4. Preferences and food-market closure

A representative household has Stone–Geary preferences over food c_a and the non-agricultural numeraire c_n, with a subsistence food requirement $\bar{c}_a > 0$:

$$U(c_a, c_n) \;=\; \alpha \ln(c_a - \bar{c}_a) + (1-\alpha)\ln c_n, \qquad \alpha \in (0,1). \tag{A8}$$

In a closed local economy where regional supply must equal regional demand ($C_a = Y_a$ and $C_n = Y_n$ across the workforce $\bar{L}$), intratemporal optimality pins the relative price of food at

$$p \;=\; \frac{\alpha}{1-\alpha} \cdot \frac{Y_n}{Y_a - \bar{c}_a \bar{L}}. \tag{A9}$$

Log-differentiating (A9) at fixed $\bar{L}$ yields the closed-economy price response:

$$\hat{p}_{\text{closed}} \;=\; \hat{Y}_n - \mathcal{M}\,\hat{Y}_a, \qquad \mathcal{M} \;\equiv\; \frac{Y_a}{Y_a - \bar{c}_a \bar{L}} \;>\; 1, \tag{A10}$$

where $\mathcal{M} > 1$ is the *subsistence multiplier*: because households must meet the subsistence floor $\bar{c}_a$, a given percentage contraction in local food output drives up the local relative food price more than one-for-one.

To nest open and closed regional economies, we introduce a reduced-form market-closure parameter $\chi \in [0,1]$:

$$\hat{p} \;=\; \chi\big(\hat{Y}_n - \mathcal{M}\,\hat{Y}_a\big), \qquad \chi \in [0,1]. \tag{A11}$$

When $\chi = 0$, the district is a small open price-taker ($\hat{p} = 0$) whose food price is pinned down by external trade arbitrage. When $\chi = 1$, the district is in autarky (A10). Intermediate values $0 < \chi < 1$ capture partial market integration, in which local supply shocks pass partially into local harvest prices.

A.2.5. Labor allocation: the seasonal sink and the cultivator margin

The total local workforce $\bar{L}$ is partitioned across four mutually exclusive Census categories whose workforce shares $\ell_j \equiv L_j/\bar{L}$ sum to one:

$$\ell_a + \ell_c + \ell_n + \ell_s \;=\; 1, \tag{A12}$$

where ℓ_a is full-time (main) agricultural wage labor, ℓ_c is self-cultivation, ℓ_n is full-time non-agricultural employment, and ℓ_s is casual *seasonal* (marginal) employment.

Workers outside self-cultivation ($\bar{L} - L_c$) draw an idiosyncratic reservation value for casual/seasonal and home work from a cumulative distribution $G(\cdot)$. A worker enters the full-time wage market ($L_a + L_n$) whenever the common formal wage w exceeds their reservation value:

$$L_a + L_n \;=\; (\bar{L} - L_c)\,G(w), \qquad L_s \;=\; (\bar{L} - L_c)\,[1 - G(w)], \qquad \eta_s \;\equiv\; \frac{w\,G'(w)}{G(w)} \;\geq\; 0, \tag{A13}$$

where η_s is the wage elasticity of full-time labor supply out of the seasonal pool.

Self-cultivation L_c is pinned down by a marginal landholder condition. Ordering own-account parcels so that the net return to self-cultivating the L_c-th marginal parcel is $\pi_c(L_c)$ with $\pi_c'(L_c) < 0$, the marginal cultivator is indifferent between working their own plot and hiring out at the

market wage: $\pi_c(L_c) = w$. A rise in the wage w raises the opportunity cost of self-cultivation and induces marginal cultivators to hire out into wage labor ($dL_c/dw = 1/\pi_c'(L_c) < 0$). Log-differentiating (A13) shows that the cultivator margin simply augments the wage elasticity of market labor supply by a positive additive constant, $\eta_{\text{eff}} = \eta_s + \frac{L_c}{\bar{L}-L_c}\,|\varepsilon_\pi|^{-1}$, where $\varepsilon_\pi \equiv L_c\pi_c'(L_c)/\pi_c(L_c) < 0$. Every proposition below holds verbatim with η_{eff} in place of η_s; we write η_s for notational simplicity and track $d\ell_c$ explicitly in Section A.8. Inframarginal family labor is held fixed, so changes in agricultural labor demand pass directly into hired agricultural wage labor $\hat{L}_a$.

A.3. The Producer's Problem

The representative farm chooses field labor L_a and cultivated land Z to maximize profit $\Pi_a = pY_a - wL_a - C(Z)$, taking p and w as given, where $C(Z)$ is a convex land cultivation cost satisfying $C'(Z) > 0$ and $C''(Z) \geq 0$.

A.3.1. The intensive margin: labor demand and factor bias

The first-order condition for agricultural labor equates the value of its marginal product to the wage, $p\,\partial Y_a/\partial L_a = w$. Differentiating (A1) with respect to L_a yields

$$\text{MPL}_a \;\equiv\; \frac{\partial Y_a}{\partial L_a} \;=\; \theta\, A^{\frac{\sigma-1}{\sigma}} \left(\frac{Y_a}{L_a}\right)^{\frac{1}{\sigma}}. \tag{A14}$$

LEMMA A1 (Factor-bias decomposition of agricultural labor demand). *To first order, the marginal product of agricultural labor satisfies*

$$\widehat{\text{MPL}}_a \;=\; \hat{A} \;+\; \frac{s_Z}{\sigma}\big(\hat{B} + \hat{Z} - \hat{L}_a\big). \tag{A15}$$

Consequently, holding cultivated area fixed ($\hat{Z} = 0$), the farm's labor demand schedule is

$$\boxed{\hat{L}_a \;=\; \hat{B} \;+\; \varepsilon_a\big(\hat{p} + \hat{A} - \hat{w}\big), \qquad \varepsilon_a \;\equiv\; \frac{\sigma}{s_Z} \;>\; 0.} \tag{A16}$$

More generally, at variable area, labor intensity per hectare $\hat{L}_a - \hat{Z}$ obeys the exact same equation: $\hat{L}_a - \hat{Z} = \hat{B} + \varepsilon_a(\hat{p} + \hat{A} - \hat{w})$.

PROOF. Taking logs of (A14) gives $\ln \text{MPL}_a = \ln\theta + \frac{\sigma-1}{\sigma}\ln A + \frac{1}{\sigma}(\ln Y_a - \ln L_a)$. Differentiating and substituting $\hat{Y}_a - \hat{L}_a = \hat{A} + s_Z(\hat{B} + \hat{Z} - \hat{L}_a)$ yields

$$\widehat{\text{MPL}}_a \;=\; \frac{\sigma-1}{\sigma}\hat{A} + \frac{1}{\sigma}\Big[\hat{A} + s_Z\big(\hat{B} + \hat{Z} - \hat{L}_a\big)\Big] \;=\; \hat{A} + \frac{s_Z}{\sigma}\big(\hat{B} + \hat{Z} - \hat{L}_a\big),$$

which establishes (A15). Substituting (A15) into the log first-order condition $\hat{p} + \widehat{\text{MPL}}_a = \hat{w}$, multiplying by $\varepsilon_a = \sigma/s_Z$, and rearranging for $\hat{L}_a - \hat{Z}$ (or $\hat{L}_a$ at $\hat{Z} = 0$) delivers (A16). □

Equation (A16) is the engine of the model. The land-biased shock $\hat{B}$ enters agricultural labor demand with a *direct unit coefficient*: a change in effective land services shifts the marginal product of labor and alters the optimal labor-to-land ratio at given prices. By contrast, the Hicks-neutral shock $\hat{A}$ enters only inside the effective profitability block $\varepsilon_a(\hat{p} + \hat{A} - \hat{w})$: it scales the entire farm operation up or down without directly altering the relative factor mix. This separation

between land-biased and neutral shocks exists if and only if the production function departs from Cobb–Douglas:

LEMMA A2 (Observational equivalence under Cobb–Douglas). *If $\sigma = 1$, the land-biased shock $\hat{B}$ is observationally equivalent to a Hicks-neutral TFP shock scaled by $s_Z = 1 - \theta$. Agricultural labor demand depends solely on the composite TFP shock $\widehat{\text{TFP}} \equiv \hat{A} + (1-\theta)\hat{B}$:*

$$\hat{L}_a \;=\; \frac{1}{1-\theta}\big(\hat{p} + \widehat{\text{TFP}} - \hat{w}\big).$$

Consequently, under $\sigma = 1$, decomposing warming into $T_{\min}$ and $T_{\max}$ carries no implication for the relative direction of labor versus output responses.

PROOF. When $\sigma = 1$, $Y_a = AL_a^{\theta}(BZ)^{1-\theta} = [AB^{1-\theta}]L_a^{\theta}Z^{1-\theta}$. Setting $\sigma = 1$ in (A16) gives $\varepsilon_a = 1/s_Z = 1/(1-\theta)$, so $\hat{L}_a = \hat{B} + \frac{1}{1-\theta}(\hat{p} + \hat{A} - \hat{w}) = \frac{1}{1-\theta}[\hat{p} + \hat{A} + (1-\theta)\hat{B} - \hat{w}]$. □

REMARK A1. *Lemma A2 explains why a Cobb–Douglas technology cannot generate opposite-signed labor responses from factor bias alone: under unit elasticity, any land-biased shock is just a reparameterized neutral TFP shock. While the algebraic sign formulas in Proposition A2 remain well-defined for any $\sigma > 0$, departing from $\sigma = 1$ is load-bearing for attributing the nighttime and daytime divergence to factor bias.*

LEMMA A3 (Equivalence under a labor-disaugmenting daytime heat shock). *Suppose daytime heat directly impairs human physiological capacity in the field, so effective agricultural labor is $A_L(T_{\max})\, L_a$ with $A_L'(T_{\max}) < 0$ and $\hat{A}_L \equiv \frac{A_L'(T_{\max})}{A_L}\, dT_{\max} \le 0$. Retaining both neutral crop stress $\hat{A} \le 0$ and worker heat stress $\hat{A}_L \le 0$, the farm's labor demand schedule becomes*

$$\hat{L}_a \;=\; \hat{B} \;+\; \varepsilon_a\big(\hat{p} + \hat{A} - \hat{w}\big) \;+\; \varepsilon_L \hat{A}_L, \qquad \varepsilon_L \;\equiv\; \frac{\sigma - s_Z}{s_Z}. \tag{A17}$$

Under Assumption A1(i) ($\sigma > 1 > s_Z$), $\varepsilon_L > 0$, so direct daytime worker heat stress ($\hat{A}_L < 0$) lowers *agricultural labor demand—reinforcing the Hicks-neutral contraction $\varepsilon_a \hat{A} < 0$.*

PROOF. With $Y_a = A\big[\theta(A_L L_a)^{\rho} + (1-\theta)(BZ)^{\rho}\big]^{1/\rho}$, the marginal product of physical labor L_a is $\text{MPL}_a = \theta(AA_L)^{\frac{\sigma-1}{\sigma}}(Y_a/L_a)^{\frac{1}{\sigma}}$. Log-differentiating and substituting $\hat{Y}_a - \hat{L}_a = \hat{A} + s_L\hat{A}_L + s_Z(\hat{B} + \hat{Z} - \hat{L}_a)$ at $\hat{Z} = 0$ gives

$$\widehat{\text{MPL}}_a \;=\; \hat{A} + \frac{\sigma - 1 + s_L}{\sigma}\hat{A}_L + \frac{s_Z}{\sigma}(\hat{B} - \hat{L}_a) \;=\; \hat{A} + \frac{\sigma - s_Z}{\sigma}\hat{A}_L + \frac{s_Z}{\sigma}(\hat{B} - \hat{L}_a),$$

using $1 - s_L = s_Z$. Substituting into $\hat{p} + \widehat{\text{MPL}}_a = \hat{w}$ and multiplying by $\varepsilon_a = \sigma/s_Z$ yields (A17). □

REMARK A2. *Whether daytime heat is modeled as a pure neutral TFP loss ($\hat{A} < 0$), a pure outdoor worker productivity loss ($\hat{A}_L < 0$), or a combination of both, it enters* (A17) *with a strictly negative effect on agricultural labor demand whenever $\sigma > s_Z$. We use the Hicks-neutral notation $\varepsilon_a\hat{A}$ as our baseline shorthand because daytime heat simultaneously stresses both crops and outdoor workers; every comparative-static sign is invariant to replacing $\varepsilon_a\hat{A}$ with $\varepsilon_a\hat{A} + \varepsilon_L\hat{A}_L$.*

REMARK A3 (Why nocturnal sleep disruption cannot replace land bias). *Equation* (A17) *also rules out an alternative hypothesis in which warmer nights operate purely by disrupting workers' sleep ($\hat{A}_L(T_{\min}) < 0$) without a land-biased channel ($\hat{B} = 0$). Whenever $\sigma > s_Z$—the exact condition under which daytime heat stress reduces farm labor demand—we have $\varepsilon_L = (\sigma - s_Z)/s_Z > 0$, so a negative nighttime worker-productivity shock $\hat{A}_L(T_{\min}) < 0$ would* reduce *agricultural labor demand. Generating*

the observed nighttime rise in agricultural wage labor under $\sigma > s_Z$ therefore requires $\hat{B} > \varepsilon_L \left|\hat{A}_L(T_{\min})\right| \geq 0$. Conversely, if one assumed strong complementarity ($\sigma < s_Z$) so that $\varepsilon_L < 0$, then daytime worker heat stress would also raise agricultural labor demand, contradicting the observed daytime exit.

A.3.2. The extensive margin: cultivated area

The first-order condition for cultivated land equates the value of the marginal product of physical land to its marginal cultivation cost, $p\,\partial Y_a/\partial Z = C'(Z)$. Log-differentiating $\mathrm{MPZ}_a = (1-\theta)(AB)^{\frac{\sigma-1}{\sigma}}(Y_a/Z)^{\frac{1}{\sigma}}$ and allowing marginal cultivation cost $C'(Z)$ to have convexity $\zeta \equiv ZC''(Z)/C'(Z) \geq 0$ plus a climate-induced cost shift $\psi_T\,dT \geq 0$ for $T \in \{T_{\min}, T_{\max}\}$ gives

$$\hat{p} + \hat{A} + \hat{B} + \frac{s_L}{\sigma}\left(\hat{L}_a - \hat{Z} - \hat{B}\right) \;=\; \zeta\,\hat{Z} + \psi_T\,dT. \tag{A18}$$

PROPOSITION A1 (Common cultivated-area contraction). *Both nighttime and daytime warming contract cultivated area:*

$$\frac{\partial \ln Z}{\partial T_{\min}} \;\approx\; \frac{\partial \ln Z}{\partial T_{\max}} \;<\; 0. \tag{A19}$$

Equality of the two semi-elasticities corresponds to a calibration of the extensive marginal-cost shifters $(\psi_{T_{\min}}, \psi_{T_{\max}})$, consistent with the empirical failure to reject equality of the area coefficients in both the decadal panel ($F = 0.01$) and the two-decade long difference ($F = 0.03$).

PROOF. Rearranging (A18) for $\hat{Z}$ gives

$$\left(\zeta + \frac{s_L}{\sigma}\right)\hat{Z} \;=\; \hat{p} + \hat{A} + \left(1 - \frac{s_L}{\sigma}\right)\hat{B} + \frac{s_L}{\sigma}\hat{L}_a - \psi_T\,dT.$$

Equivalently, substituting $\hat{L}_a - \hat{Z} - \hat{B} = \varepsilon_a(\hat{p} + \hat{A} - \hat{w})$ from Lemma A1 directly into (A18) yields the factor-price form

$$\zeta\,\hat{Z} \;=\; \frac{1}{s_Z}\left(\hat{p} + \hat{A} - s_L\hat{w}\right) + \hat{B} - \psi_T\,dT.$$

Under daytime warming ($T_{\max}$), $\hat{A} < 0$ and $-\psi_{T_{\max}} dT_{\max} < 0$ both depress the right-hand side; in the empirically relevant partial-open regime ($\chi < \chi^*$, Proposition A5), the endogenous price response $\hat{p} > 0$ is bounded below $\left|\hat{A}\right|$, so $\hat{Z} < 0$. Under nighttime warming ($T_{\min}$), $\hat{p} \approx 0$ and $\hat{A} = 0$, so whenever the extensive marginal-land stress $\psi_{T_{\min}} dT_{\min}$ exceeds the intensive term $\hat{B} - \frac{s_L}{s_Z}\hat{w}$, cultivated area contracts ($\hat{Z} < 0$). Matching the two semi-elasticities is a calibration of $(\psi_{T_{\min}}, \psi_{T_{\max}})$, equivalent in the mean-and-range decomposition to the area contraction loading purely on the diurnal mean. □

REMARK A4 (Area contraction is common; labor reallocation is intensive). *Proposition A1 captures the symmetric contraction of cultivated land across nighttime and daytime warming. Because Lemma A1 establishes that labor intensity on cultivated land, $\hat{L}_a - \hat{Z} = \hat{B} + \varepsilon_a(\hat{p} + \hat{A} - \hat{w})$, is independent of $\hat{Z}$, the opposite-signed divergence in worker shares is governed by the intensive factor-bias block. We therefore solve the general-equilibrium labor reallocation block at fixed area ($\hat{Z} = 0$) and treat extensive land abandonment as the common level effect.*

REMARK A5 (Worker-status shares versus effective field-labor hours). *The Census measures a worker's primary occupational* status *($L_a/\bar{L}$), which maps into the* sign *of effective field labor $\hat{L}_a$ rather than moving total field-hours one-for-one in levels. If the Census status share moved effective field-hours*

one-for-one in levels, then substituting the open-economy labor response $\hat{L}_a = \tau_a(\hat{B} + \varepsilon_a \hat{A})$ *with* $\tau_a \in (0,1)$ *into the yield identity* $\hat{y}_a = \hat{A} + s_L(\hat{L}_a - \hat{Z}) + s_Z \hat{B}$ *under* $\sigma = \varepsilon_a s_Z > 1$ *would imply*

$$\hat{y}_{a,T_{\min}} \;=\; s_L \hat{L}_{a,T_{\min}} + s_L \left|\hat{Z}\right| + \frac{s_Z}{\tau_a} \hat{L}_{a,T_{\min}} \;>\; (s_L + s_Z)\hat{L}_{a,T_{\min}} \;=\; \hat{L}_{a,T_{\min}},$$

requiring the log yield response at night to exceed the log rise in agricultural wage workers. In the data, the log rise in agricultural wage laborers ($\approx +0.24$ *on an* 0.18 *base) is larger than the dry-bulb log yield response (*$+0.052$ *in the long difference), because shifts from seasonal or self-cultivating status into main wage-labor status reflect contractual reorganization and task reallocation rather than a proportional jump in total biological field hours. The theoretical framework therefore disciplines* signs *and* relative orderings *across margins rather than imposing cross-equation level identities.*

A.4. Closing the Economy

Collecting the log-linearized equilibrium conditions for the intensive labor and price block gives a five-equation system:

(A20) (agricultural labor demand) $\hat{L}_a \;=\; \hat{B} + \varepsilon_a\big(\hat{p} + \hat{A} - \hat{w}\big),$

(A21) (non-agricultural labor demand) $\hat{L}_n \;=\; \varepsilon_n\big(\hat{A}_n - \hat{w}\big), \qquad \varepsilon_n \;\equiv\; \dfrac{1}{1-\nu},$

(A22) (market labor clearing) $\phi_a \hat{L}_a + \phi_n \hat{L}_n \;=\; \eta_s\, \hat{w},$

(A23) (food-market closure) $\hat{p} \;=\; \chi\big(\hat{Y}_n - \mathcal{M}\, \hat{Y}_a\big),$

(A24) (sectoral outputs) $\hat{Y}_a \;=\; \hat{A} + s_L \hat{L}_a + s_Z \hat{B}, \qquad \hat{Y}_n \;=\; \hat{A}_n + \nu\, \hat{L}_n,$

where $\phi_a \equiv L_a/(L_a + L_n)$ and $\phi_n \equiv 1 - \phi_a$ are the initial shares of full-time market labor.

For the baseline district analysis ($\hat{A}_n = 0$), substituting (A21) into (A22) immediately yields the inverse labor supply schedule facing the agricultural sector:

(A25) $$\phi_a \hat{L}_a \;=\; \eta_a\, \hat{w} \quad\Longleftrightarrow\quad \hat{w} \;=\; \frac{\phi_a}{\eta_a}\, \hat{L}_a, \qquad \text{where} \quad \eta_a \;\equiv\; \eta_s + \phi_n \varepsilon_n \;>\; 0.$$

Economically, η_a is the *effective labor supply elasticity facing agriculture*: each percent rise in the local wage draws workers out of the seasonal pool (with elasticity η_s) and out of non-agriculture (with weighted elasticity $\phi_n \varepsilon_n$).

A.5. The Open Economy

In a small open district economy ($\chi = 0$), trade arbitrage pins the relative food price at the external benchmark ($\hat{p} = 0$). Substituting $\hat{w} = (\phi_a/\eta_a)\hat{L}_a$ and $\hat{p} = 0$ into (A20) yields the open-economy equilibrium:

(A26) $$\hat{L}_a \;=\; \tau_a\big(\hat{B} + \varepsilon_a \hat{A}\big), \qquad \hat{w} \;=\; \frac{\phi_a}{\eta_a}\, \tau_a\big(\hat{B} + \varepsilon_a \hat{A}\big), \qquad \text{where} \quad \tau_a \;\equiv\; \frac{\eta_a}{\eta_a + \phi_a \varepsilon_a} \;\in\; (0,1).$$

Here $\tau_a \in (0,1)$ is the standard Marshallian incidence ratio: the ratio of the effective labor supply elasticity η_a to the sum of supply and weighted demand elasticities $\eta_a + \phi_a \varepsilon_a$.

PROPOSITION A2 (Diurnal labor divergence in the open economy). *Under Assumption A1, agricultural wage labor rises with nighttime warming and falls with daytime warming:*

$$\frac{\partial \hat{L}_a}{\partial T_{\min}} \;=\; \tau_a \cdot \frac{B'(T_{\min})}{B} \;\geq\; 0, \qquad \frac{\partial \hat{L}_a}{\partial T_{\max}} \;=\; \tau_a\, \varepsilon_a \cdot \frac{A'(T_{\max})}{A} \;<\; 0, \tag{A27}$$

with strictly opposite signs whenever $B'(T_{\min}) > 0$ and $A'(T_{\max}) < 0$. The seasonal labor pool moves oppositely to agricultural wage labor ($\hat{L}_s \propto -\hat{w} \propto -\hat{L}_a$).

PROOF. Differentiating (A26) with respect to $T_{\min}$ ($\hat{B} = (B'/B)dT_{\min}, \hat{A} = 0$) and $T_{\max}$ ($\hat{A} = (A'/A)dT_{\max}, \hat{B} = 0$) gives the stated expressions since $\tau_a > 0$ and $\varepsilon_a > 0$. From (A13), the seasonal pool response $\hat{L}_s = -\frac{L_a + L_n}{L_s}\eta_s \hat{w}$ has the sign of $-\hat{w} = -\frac{\phi_a}{\eta_a}\hat{L}_a$, moving opposite to $\hat{L}_a$. □

REMARK A6. *Proposition A2 is the diurnal analogue of the factor-bias theorem in Bustos, Caprettini, and Ponticelli (2016): land-augmenting shocks raise farm labor demand in an open economy, whereas neutral (or labor-disaugmenting) negative shocks reduce it. Moreover, in the mean-and-range reparameterization ($T_{\min} = T_{\text{mean}} - \frac{1}{2}\text{DTR}$, $T_{\max} = T_{\text{mean}} + \frac{1}{2}\text{DTR}$), the opposite-signed night and day effects offset on T_{mean} ($\beta_{\text{mean}} = \beta_{T_{\min}} + \beta_{T_{\max}} \approx 0$) and load onto the diurnal range ($\beta_{\text{DTR}} = \frac{1}{2}(\beta_{T_{\max}} - \beta_{T_{\min}}) < 0$), matching the long-difference rural range estimate ($-0.082, t = -5.6$).*

PROPOSITION A3 (Decoupling of labor and yield under nighttime warming). *In the open economy:*

(i) The sign of the nighttime agricultural labor response is pinned down solely by the sign of the land-bias shock, $\text{sgn}(\partial \hat{L}_a / \partial T_{\min}) = \text{sgn}(B'(T_{\min}))$*, with attenuation $\tau_a \in (0,1)$ governed by labor-market elasticities;*

(ii) The crop yield response $\hat{y}_a = \hat{A} + s_L(\hat{L}_a - \hat{Z}) + s_Z \hat{B}$ is weakly positive under nighttime warming ($\hat{A} = 0, \hat{B} \geq 0, \hat{L}_a > 0, \hat{Z} \leq 0$) and strictly exceeds the daytime yield response whenever $B'(T_{\min}) \geq 0 > A'(T_{\max})$;

(iii) Under daytime warming ($\hat{B} = 0, \hat{A} < 0$), labor, output, and yield all move in the same negative direction: $\text{sgn}(\partial \hat{L}_a / \partial T_{\max}) = \text{sgn}(\partial \hat{Y}_a / \partial T_{\max}) = \text{sgn}(A'(T_{\max})) < 0$.

PROOF. Parts (i) and (iii) follow directly from Proposition A2 and $\hat{Y}_a = \hat{A} + s_L \hat{L}_a + s_Z(\hat{B} + \hat{Z})$. For part (ii), substituting $\hat{L}_a = \tau_a \hat{B} \geq 0$ and $\hat{Z} \leq 0$ under $T_{\min}$ ($\hat{A} = 0$) gives $\hat{y}_{a,T_{\min}} = s_L(\tau_a \hat{B} - \hat{Z}) + s_Z \hat{B} \geq 0$, whereas under $T_{\max}$ ($\hat{B} = 0, \hat{A} < 0$), intense labor shedding ($\hat{L}_{a,T_{\max}} = \tau_a \varepsilon_a \hat{A} < 0$) and direct TFP loss ($\hat{A} < 0$) pull $\hat{y}_{a,T_{\max}}$ downward, generating a positive night-minus-day yield wedge ($\hat{y}_{a,T_{\min}} > \hat{y}_{a,T_{\max}}$). □

REMARK A7 (Empirical evidence on the land-bias sign and yield divergence). *Proposition A3(ii) predicts both a positive night-minus-day yield divergence and, when dry-heat evaporative stress on marginal land is netted out, an outright positive nighttime yield effect. Both predictions are borne out in the data:*

1. *In the* two-decade dry-bulb long difference, *the nighttime cereal yield estimate is slightly positive ($+0.052$) while the daytime estimate is negative (-0.178), rejecting equality at $F = 4.56$ ($p = 0.034$, equivalent to a diurnal-range yield t-statistic of -2.14).*
2. *Under* humidity-adjusted (wet-bulb) temperatures, *which net out dry-air evaporative stress, nighttime warming raises cereal yields outright in both the decadal panel ($+0.15, p = 0.049$) and the long difference ($+0.36, p = 0.03$) without contracting cultivated area. Because wet-bulb $T_{\min}$ and $T_{\max}$ are strongly collinear after fixed effects* (VIF ≈ 3.5 *vs.* VIF ≈ 1.0 *in dry-bulb), we treat the dry-bulb estimates as our cleanly identified baseline and the wet-bulb yield dividend as corroborative evidence of $B'(T_{\min}) > 0$.*

REMARK A8 (Bounding non-agricultural productivity spillovers in rural districts). *Could the nighttime rise in rural agricultural labor be driven by a negative nighttime shock to rural non-agricultural productivity ($\hat{A}_n(T_{\min}) < 0$) rather than land bias ($\hat{B} > 0$)? Equation* (A35) *in Appendix A.12 shows that a $\hat{A}_n < 0$ shock raises $\hat{L}_a$ only by pushing workers* out *of district non-agriculture ($\hat{L}_n < 0$). In the data, district non-agricultural worker shares do not contract under nighttime warming ($+0.017, t = 1.46$ in the panel; $+0.026, t = 0.94$ in the long difference), and the nighttime inflow into agricultural wage labor is supplied entirely by self-cultivators and the seasonal pool (Table 2). Thus $\hat{A}_n$ cannot account for the rural nighttime agricultural labor rise.*

COROLLARY A1 (Open-economy wages and prices). *At $\chi = 0$, local harvest prices do not respond to local climate shocks ($\hat{p} = 0$). By* (A26)*, the field wage falls under daytime warming ($\partial\hat{w}/\partial T_{\max} < 0$) and is weakly positive under nighttime warming ($\partial\hat{w}/\partial T_{\min} \geq 0$, scaled down by ϕ_a/η_a). Generating a positive local harvest-price response to daytime warming requires partial local market closure ($\chi > 0$).*

A.6. The Closed Economy and the Openness Threshold

Now allow endogenous local food-price adjustments ($\chi \in (0, 1]$). Substituting $\hat{Y}_n = -\frac{\nu}{1-\nu}\hat{w} = -\frac{\nu}{1-\nu}\frac{\phi_a}{\eta_a}\hat{L}_a$ and $\hat{Y}_a = \hat{A} + s_L\hat{L}_a + s_Z\hat{B}$ into the price closure (A23), and inserting the resulting $\hat{p}$ into agricultural labor demand (A20) using $\varepsilon_a s_Z = \sigma$ (full derivation in Appendix A.12), we obtain the general equilibrium solution for any $\chi \in [0, 1]$:

$$\boxed{\hat{L}_a \;=\; \frac{1 - \sigma\chi\mathcal{M}}{D(\chi)}\,\hat{B} \;+\; \frac{\varepsilon_a(1 - \chi\mathcal{M})}{D(\chi)}\,\hat{A},} \tag{A28}$$

where the denominator is strictly positive:

$$D(\chi) \;\equiv\; 1 \;+\; \varepsilon_a\chi\mathcal{M}s_L \;+\; \frac{\phi_a\varepsilon_a}{\eta_a}\Big(1 + \chi\frac{\nu}{1-\nu}\Big) \;>\; 0. \tag{A29}$$

Setting $\chi = 0$ immediately recovers the open-economy solution (A26) since $1/D(0) = \frac{\eta_a}{\eta_a + \phi_a\varepsilon_a} = \tau_a$. Setting $\chi = 1$ yields the fully closed subsistence economy.

PROPOSITION A4 (Sign reversal in the closed subsistence economy). *In a fully closed subsistence economy ($\chi = 1$) with $\mathcal{M} > 1$ and $\sigma > 1$, both diurnal labor signs reverse:*

$$\frac{\partial\hat{L}_a}{\partial T_{\min}} \;<\; 0 \qquad \textit{and} \qquad \frac{\partial\hat{L}_a}{\partial T_{\max}} \;>\; 0. \tag{A30}$$

Nighttime warming lowers *and daytime warming* raises *agricultural labor—the exact opposite of the open economy and of the Indian data.*

PROOF. Evaluate (A28) at $\chi = 1$. Since $\mathcal{M} > 1$ and $\sigma > 1$, we have $1 - \chi\mathcal{M} = 1 - \mathcal{M} < 0$ and $1 - \sigma\chi\mathcal{M} = 1 - \sigma\mathcal{M} < 0$. With $A'(T_{\max}) < 0$ and $B'(T_{\min}) > 0$, it follows that $\partial\hat{L}_a/\partial T_{\max} = \frac{\varepsilon_a(1-\mathcal{M})}{D(1)}\frac{A'}{A} > 0$ and $\partial\hat{L}_a/\partial T_{\min} = \frac{1-\sigma\mathcal{M}}{D(1)}\frac{B'}{B} < 0$. Economically, when food cannot be traded across districts, a daytime productivity loss triggers a severe subsistence price spike ($\mathcal{M} > 1$) that forces labor *into* agriculture to meet minimum food needs, while a land-augmenting nighttime shock saturates local food demand and depresses p enough to release labor. □

PROPOSITION A5 (Openness threshold for the observed sign orientation). *Under Assumption A1 with $B'(T_{\min}) > 0$, the equilibrium exhibits the empirically observed orientation—nighttime warming*

raising ($\partial \hat{L}_a / \partial T_{\min} > 0$) and daytime warming lowering ($\partial \hat{L}_a / \partial T_{\max} < 0$) agricultural wage labor—if and only if local food-market closure lies below the threshold

$$\chi \;<\; \chi^{*} \;\equiv\; \frac{1}{\sigma \mathcal{M}}. \tag{A31}$$

Because $\sigma > 1$, the nighttime condition ($\chi < \frac{1}{\sigma \mathcal{M}}$) is strictly more binding than the daytime condition ($\chi < \frac{1}{\mathcal{M}}$).

Proof. From (A28), $\partial \hat{L}_a / \partial T_{\max} < 0$ holds iff $1 - \chi \mathcal{M} > 0 \iff \chi < 1/\mathcal{M}$, whereas $\partial \hat{L}_a / \partial T_{\min} > 0$ holds iff $1 - \sigma \chi \mathcal{M} > 0 \iff \chi < 1/(\sigma \mathcal{M})$. Since $\sigma > 1$, the intersection is $\chi < \chi^{*} \equiv 1/(\sigma \mathcal{M})$. □

remark A9 (Market integration across time horizons). *Proposition A5 shows that whenever nighttime warming raises agricultural labor, the economy must lie in the integrated regime $\chi < \chi^{*}$. Because $\partial \hat{L}_a / \partial T_{\min} > 0$ holds in both rural and urban district populations across both the decadal panel and the two-decade long difference, Indian districts operate well inside the open threshold ($\chi < \chi^{*}$). Within $0 \le \chi < \chi^{*}$, the decadal panel reflects small positive local closure ($0 < \chi < \chi^{*}$), generating the medium-run daytime harvest-price spike ($F = 13.34$), whereas over the two-decade long difference local harvest prices do not significantly diverge ($F = 1.81$), consistent with long-run price arbitrage approaching the $\chi \to 0$ benchmark.*

A.7. Comparative Statics for the Four Agricultural Margins

We summarize the model's predictions for the four ICRISAT agricultural outcomes in the empirically selected regime $0 \le \chi < \chi^{*}$:

1. **Cultivated area ($\hat{Z}$):** Contracts equally under both $T_{\min}$ and $T_{\max}$ (Proposition A1), loading on the diurnal mean.
2. **Crop yield ($\hat{y}_a = \hat{Y}_a - \hat{Z}$):** Rises (or holds flat) under nighttime warming as hired labor intensifies on remaining land ($\hat{L}_a - \hat{Z} > 0$), and falls under daytime warming as both TFP and labor retreat ($\hat{y}_{a,T_{\min}} > \hat{y}_{a,T_{\max}}$, loading negatively on the diurnal range; confirmed in the long difference at $F = 4.56$ and in wet-bulb temperatures).
3. **Field wage ($\hat{w} = \frac{\phi_a}{\eta_a} \hat{L}_a$):** Falls under daytime warming as neutral TFP loss depresses labor demand; under nighttime warming the theoretical open-economy wage effect is weakly positive but small (attenuated by ϕ_a / η_a), while empirically the night and day wage estimates do not statistically separate ($F = 0.74$ in the panel; $F = 0.02$ in the long difference).
4. **Harvest price ($\hat{p} = \chi(\hat{Y}_n - \mathcal{M} \hat{Y}_a)$):** When $\chi > 0$ (medium-run decadal panel), daytime warming raises the local harvest price ($\partial \hat{p} / \partial T_{\max} > 0$, $F = 13.34$), whereas nighttime warming leaves it flat; as $\chi \to 0$ over the two-decade long difference, the local price wedge attenuates ($F = 1.81$).

remark A10 (Marketed surplus versus subsistence marginal parcels). *Why does nighttime cultivated-area abandonment in the decadal panel not raise the local harvest price alongside daytime warming? In Section A.2.5, nighttime land abandonment is concentrated on marginal self-cultivated plots whose output is consumed on-farm: when a marginal cultivator gives up their plot and hires out as a wage laborer, their household's subsistence demand and supply move together, leaving* marketed surplus *on commercial inframarginal land buffered by higher hired-labor intensity. By contrast, daytime heat reduces productivity across all inframarginal marketed grain, tightening local marketed supply and raising the harvest price whenever $\chi > 0$. Under wet-bulb temperatures, where nighttime warming raises inframarginal yields*

outright without area loss, marketed supply expands and the nighttime harvest price falls significantly ($-0.237, F = 16.59$; Table A14), confirming that the price closure operates on marketed surplus.

PROPOSITION A6 (Summary of the four agricultural margins). *In the regime $0 \leq \chi < \chi^*$ under Assumption A1:*

	Cultivated area ($\hat{Z}$)	Yield ($\hat{y}_a$)	Field wage ($\hat{w}$)	Harvest price ($\hat{p}$)
$T_{\min}$ *(night)*	$(-)$	$(\approx 0, +)$	(≈ 0)	(≈ 0)
$T_{\max}$ *(day)*	$(-)$	$(\approx 0, -)$	$(-)$	$(+$ *if* $\chi > 0$; ≈ 0 *if* $\chi \to 0)$

PROOF. Collect Proposition A1 (area), Proposition A3(ii)–(iii) (yield), Corollary A1 (field wage), and (A23) with Remark A10 (harvest price). □

A.8. Within-Agriculture Workforce Composition

Differentiating the workforce adding-up identity $\ell_a + \ell_c + \ell_n + \ell_s = 1$ in (A12) connects the agricultural wage-labor elasticity $\hat{L}_a$ to the four observed Census shares.

LEMMA A4 (Counter-flows between wage labor, self-cultivation, and seasonal work). *When district non-agricultural employment shares remain flat ($d\ell_n \approx 0$), changes in the agricultural wage-labor share $d\ell_a = \ell_a \hat{L}_a$ are exactly offset by the sum of self-cultivation and seasonal worker shares:*

$$d\ell_c + d\ell_s \;=\; -\,d\ell_a \;=\; -\,\ell_a\,\hat{L}_a. \tag{A32}$$

Under nighttime warming ($\hat{L}_a > 0$), $d\ell_c + d\ell_s < 0$: expanding farm wage labor draws workers out of both marginal self-cultivation ($d\ell_c < 0$) and the seasonal pool ($d\ell_s < 0$). Under daytime warming ($\hat{L}_a < 0$), $d\ell_c + d\ell_s > 0$: displaced agricultural wage workers are absorbed primarily by the elastic seasonal buffer ($d\ell_s > 0$).

PROOF. Setting $d\ell_n = 0$ in $d\ell_a + d\ell_c + d\ell_n + d\ell_s = 0$ gives $d\ell_c + d\ell_s = -d\ell_a$. The signs of $d\ell_a$ follow from Proposition A2, and the division between $d\ell_c$ and $d\ell_s$ reflects the high elasticity η_s of the casual seasonal pool when farms shed wage labor. □

A.9. Rural versus Urban Populations: Amplitude

Because both rural and urban district populations satisfy $\chi < \chi^*$ (Remark A9), they share the same theoretical sign structure. Their striking empirical contrast in Figure 2 ($F = 30.9$ in rural populations vs. $F = 1.9$ in urban populations) is an *amplitude* difference governed by two structural parameters.

PROPOSITION A7 (Rural amplification of the diurnal labor divergence). *For a given climate shock, the change in the agricultural wage-labor share $d\ell_a = \ell_a \hat{L}_a$ is strictly larger in magnitude in rural populations than in urban populations:*

$$\left| d\ell_a^{\text{rural}} \right| \;>\; \left| d\ell_a^{\text{urban}} \right|, \tag{A33}$$

driven by two reinforcing channels:

(i) ***Baseline agricultural scale (ℓ_a):*** *The initial agricultural wage-labor share ℓ_a is several times larger in rural than in urban populations, scaling any given percentage response $\hat{L}_a$ into a proportionally larger percentage-point share change $d\ell_a$;*

(ii) ***Depth of the seasonal labor buffer (η_s):*** *The casual seasonal supply elasticity η_s is deeper in rural areas, which increases $\eta_a = \eta_s + \phi_n \varepsilon_n$ and raises the pass-through coefficient $\tau_a = \frac{\eta_a}{\eta_a + \phi_a \varepsilon_a}$ toward 1:*

$$\frac{\partial \tau_a}{\partial \eta_s} \;=\; \frac{\phi_a \varepsilon_a}{(\eta_a + \phi_a \varepsilon_a)^2} \;>\; 0.$$

PROOF. From (A26), $d\ell_a = \ell_a \tau_a (\hat{B} + \varepsilon_a \hat{A})$. Channel (i) is immediate from $\partial \left| d\ell_a \right| / \partial \ell_a = \left| \hat{L}_a \right| > 0$, which dominates the bounded feedback of ϕ_a in $\tau_a \in (0, 1)$. Channel (ii) follows by differentiating $\tau_a = \eta_a / (\eta_a + \phi_a \varepsilon_a)$ with respect to η_s. □

REMARK A11 (Market closure dampens rather than amplifies the divergence). *Notice from* (A28) *that on $\chi \in [0, \chi^*)$, both numerators $1 - \sigma\chi\mathcal{M}$ and $\varepsilon_a(1 - \chi\mathcal{M})$ decrease in χ while the denominator $D(\chi)$ increases in χ. Greater local market closure χ therefore* dampens *the labor divergence toward zero. If rural areas were simply more closed than urban areas, the rural divergence would be smaller, not larger; hence the rural amplification operates through ℓ_a and η_s.*

A.10. Census Towns and the Non-Agricultural Drag

In Census towns, where by construction over 75% of male workers are employed outside agriculture (ℓ_a and ϕ_a are very small), Proposition A7 implies that the within-agriculture factor-bias divergence is heavily attenuated. Instead, the dominant margin in towns is the direct thermal sensitivity of urban informal non-agricultural work, $\hat{A}_n < 0$ under both daytime heat strain and nighttime sleep/recovery loss.

PROPOSITION A8 (Unilateral non-agricultural contraction in towns). *Suppose non-agricultural productivity falls with both nighttime and daytime warming, $\partial \hat{A}_n / \partial T_{\min} < 0$ and $\partial \hat{A}_n / \partial T_{\max} < 0$. Then in an open town economy ($\chi \approx 0$) where agriculture is a minor share of employment (ϕ_a small), non-agricultural employment contracts under both margins:*

$$\frac{\partial \hat{L}_n}{\partial T_{\min}} \;<\; 0 \qquad \textit{and} \qquad \frac{\partial \hat{L}_n}{\partial T_{\max}} \;<\; 0. \tag{A34}$$

PROOF. Solving (A20)–(A22) at $\chi = 0$ with $\hat{A}_n \neq 0$ (derived in Appendix A.12, equation (A35)) gives

$$\hat{L}_n \;=\; \frac{\varepsilon_n(\eta_s + \phi_a \varepsilon_a)}{\eta_a + \phi_a \varepsilon_a}\, \hat{A}_n \;-\; \frac{\phi_a \varepsilon_n}{\eta_a + \phi_a \varepsilon_a} \left(\hat{B} + \varepsilon_a \hat{A} \right).$$

Because $\varepsilon_n(\eta_s + \phi_a \varepsilon_a) / (\eta_a + \phi_a \varepsilon_a) > 0$, the direct effect of $\hat{A}_n < 0$ is strictly negative under both $T_{\min}$ and $T_{\max}$. With the town agricultural employment share ϕ_a small, the second term is small and cannot overturn the direct non-agricultural contraction. □

REMARK A12 (Peri-urban agricultural retention in southern towns). *While Proposition A8 accounts for the bilateral contraction in town non-agricultural worker shares ($\Delta T_{\min}$: -0.038, $\Delta T_{\max}$: -0.078, $F = 2.78$), in peninsular-south towns warming also raises the peri-urban agricultural wage-labor share, erasing its secular decline. Because the log-linear model attenuates agricultural share shifts when baseline ℓ_a is small, we treat this regional peri-urban retention as an empirical regularity alongside the modeled non-agricultural drag.*

A.11. Load-Bearing Primitives and Mapping to the Estimates

Table A1 summarizes each theoretical prediction, the primitive assumption that carries it, and its empirical counterpart in the main text and appendix.

Prediction (result)	Load-bearing primitive	Empirical counterpart
Land-biased night shock is distinct from a neutral day shock (Lm. A1, A2)	$\sigma \neq 1$; $\sigma > 1$ maintained	Cobb–Douglas ($\sigma = 1$) folds $\hat{B}$ into TFP and leaves no diurnal labor divergence.
Nights raise, days lower agricultural wage labor (Pr. A2, A3)	$A' < 0$, $B' > 0$, $\chi < \chi^*$	Ag. labor share: night $(+)$, day $(-)$; $F = 24.9$ in panel (Table 2), $F = 26.1$ in LD (Table A8), loading on DTR.
Wage labor drawn from cultivation and seasonal pools (nights), shed to seasonal (days) (Lm. A4)	Workforce adding-up $\sum_j \ell_j = 1$ with $d\ell_n \approx 0$	Cultivator share drawn off at night; seasonal share moves opposite to wage labor, $F = 14.7$ in panel, $F = 12.3$ in LD.
Cultivated area contracts equally under both margins (Pr. A1)	Extensive cost shifters ψ_T; equality is calibration	Equal area contraction: $F = 0.01$ in panel (Table 1), $F = 0.03$ in LD, loading on diurnal mean.
Daytime warming raises harvest price in medium run; attenuates in long run (Pr. A6, Rem. A10)	$0 < \chi < \chi^*$ in medium run; $\chi \to 0$ in long run	Panel harvest price equality rejected, $F = 13.34$ (Table 1); LD price wedge insignificant, $F = 1.81$.
Daytime warming lowers field wage; pair does not separate (Pr. A6)	$\hat{A} < 0$; $\hat{w} = (\phi_a/\eta_a)\hat{L}_a$	Day wage negative (-0.12 panel, -0.21 LD); pair does not separate ($F = 0.74$ panel, $F = 0.02$ LD).
Nighttime yield exceeds daytime yield; positive night yield isolated in wet-bulb (Pr. A3, Rem. A7)	Land bias of $T_{\min}$ ($B' > 0$) and labor intensification	Dry-bulb LD yield divergence ($+0.052$ vs. -0.178, $F = 4.56$); wet-bulb night yield $+0.15$ (panel) and $+0.36$ (LD) (Table A14).
Rural amplifies, urban mutes; same sign structure (Pr. A7)	Rural scale ℓ_a and seasonal depth η_s, not closure χ	Ag. labor $F = 30.9$ (rural) vs. 1.9 (urban) in LD (Figure 2; Table A8).
Closed subsistence economy would reverse both labor signs (Pr. A4, A5)	$\mathcal{M} > 1$, $\sigma > 1 \implies \chi^* = \frac{1}{\sigma\mathcal{M}}$	Reversal rejected in both populations $\Rightarrow$ districts operate in $\chi < \chi^*$.
Towns: both margins depress non-agricultural employment (Pr. A8)	$\hat{A}_n < 0$ on both margins; small town ϕ_a	Town non-ag. share -0.038 (night) and -0.078 (day), $F = 2.78$ (Figure 3; Table A9).

TABLE A1. Model predictions, load-bearing primitives, and empirical counterparts

Notes: Signs are the framework's predictions in the empirically selected regime $0 \leq \chi < \chi^*$ under gross substitutes ($\sigma > 1$). Only signs and relative orderings are predicted, not levels. Table and figure numbers without an "A" refer to the main text.

A.12. Closed- and Open-Economy Algebra

Derivation of the general closed/open solution (A28). With $\hat{A}_n = 0$, equations (A21) and (A22) give $\hat{L}_n = -\varepsilon_n \hat{w}$ and $\hat{w} = (\phi_a/\eta_a)\hat{L}_a$, where $\eta_a \equiv \eta_s + \phi_n \varepsilon_n$ and $\varepsilon_n \equiv 1/(1-\nu)$. Sectoral output responses in (A24) are $\hat{Y}_n = \nu \hat{L}_n = -\frac{\nu}{1-\nu}\hat{w}$ and $\hat{Y}_a = \hat{A} + s_L \hat{L}_a + s_Z \hat{B}$. Substituting these into the price closure (A23) yields

$$\hat{p} \;=\; \chi\Big[-\frac{\nu}{1-\nu}\hat{w} - \mathcal{M}\big(\hat{A} + s_L \hat{L}_a + s_Z \hat{B}\big)\Big].$$

Inserting $\hat{p}$ into the agricultural labor demand schedule (A20), $\hat{L}_a = \hat{B} + \varepsilon_a(\hat{p} + \hat{A} - \hat{w})$, gives

$$\hat{L}_a \;=\; \hat{B} + \varepsilon_a\Big[(1-\chi\mathcal{M})\hat{A} - \chi\mathcal{M}s_Z\hat{B} - \chi\mathcal{M}s_L\hat{L}_a - \Big(1+\chi\frac{\nu}{1-\nu}\Big)\hat{w}\Big].$$

Using the identity $\varepsilon_a s_Z = \frac{\sigma}{s_Z}s_Z = \sigma$ and substituting $\hat{w} = (\phi_a/\eta_a)\hat{L}_a$, we collect all $\hat{L}_a$ terms on the left-hand side:

$$\Big[1 + \varepsilon_a\chi\mathcal{M}s_L + \frac{\phi_a\varepsilon_a}{\eta_a}\Big(1+\chi\frac{\nu}{1-\nu}\Big)\Big]\hat{L}_a \;=\; (1-\sigma\chi\mathcal{M})\hat{B} + \varepsilon_a(1-\chi\mathcal{M})\hat{A}.$$

Defining $D(\chi) \equiv 1 + \varepsilon_a\chi\mathcal{M}s_L + \frac{\phi_a\varepsilon_a}{\eta_a}\big(1+\chi\frac{\nu}{1-\nu}\big) > 0$ and dividing by $D(\chi)$ delivers (A28). At $\chi = 0$, $D(0) = 1 + \frac{\phi_a\varepsilon_a}{\eta_a} = \frac{\eta_a+\phi_a\varepsilon_a}{\eta_a} = \tau_a^{-1}$, recovering (A26).

Open economy with non-agricultural productivity shocks ($\hat{A}_n \neq 0$). Retaining $\hat{A}_n \neq 0$ in (A21) at $\chi = 0$ ($\hat{p} = 0$), substitute $\hat{L}_a = \hat{B} + \varepsilon_a(\hat{A} - \hat{w})$ and $\hat{L}_n = \varepsilon_n(\hat{A}_n - \hat{w})$ into market labor clearing $\phi_a\hat{L}_a + \phi_n\hat{L}_n = \eta_s\hat{w}$:

$$\phi_a\big(\hat{B} + \varepsilon_a\hat{A}\big) + \phi_n\varepsilon_n\hat{A}_n \;=\; \big(\eta_s + \phi_n\varepsilon_n + \phi_a\varepsilon_a\big)\hat{w} \;=\; \big(\eta_a + \phi_a\varepsilon_a\big)\hat{w}.$$

Solving for $\hat{w}$ and substituting back into $\hat{L}_a$ and $\hat{L}_n$ yields:

(A35)
$$\begin{aligned}
\hat{w} \;&=\; \frac{\phi_a(\hat{B} + \varepsilon_a\hat{A}) + \phi_n\varepsilon_n\hat{A}_n}{\eta_a + \phi_a\varepsilon_a},\\
\hat{L}_a \;&=\; \tau_a\big(\hat{B} + \varepsilon_a\hat{A}\big) \;-\; \frac{\phi_n\varepsilon_a\varepsilon_n}{\eta_a + \phi_a\varepsilon_a}\,\hat{A}_n,\\
\hat{L}_n \;&=\; \frac{\varepsilon_n(\eta_s + \phi_a\varepsilon_a)}{\eta_a + \phi_a\varepsilon_a}\,\hat{A}_n \;-\; \frac{\phi_a\varepsilon_n}{\eta_a + \phi_a\varepsilon_a}\big(\hat{B} + \varepsilon_a\hat{A}\big).
\end{aligned}$$

These expressions undergird Remark A8 and Proposition A8.

Appendix B. Identification and Inference

This section derives the panel estimator used throughout the paper, the two standard-error families it is paired with (district clustering and Conley spatial–temporal HAC), and the simulation that selects the inference procedure by its empirical *size*. It then explains why the estimator's state-specific linear trends are economically, not merely statistically, motivated, and reports the diagnostics on the diurnal regressors themselves.

Caveats, stated upfront. The size simulation below is a deliberately narrow diagnostic. Four limitations bound what it can establish:

1. **Size, not power or consistency.** The exercise asks only how often a procedure rejects a null that is true by construction. It speaks to the validity of inference (the standard errors), not to the consistency or efficiency of the point estimator.
2. **The placebo is spatially blind.** The data-generating process (DGP) permutes the *observed* climate series across spatial units. This preserves each series' decadal persistence and marginal distribution but *destroys* the cross-unit spatial correlation of climate, so the simulation stresses only the serial dimension of dependence (§B.4).
3. **Asymptotic Conley inference.** The Conley standard errors are computed on the partialled-out system, so the degrees of freedom are not adjusted for the absorbed fixed effects and trends; inference is large-sample (normal) and, if anything, mildly anti-conservative.
4. **Short decadal panel.** The estimation panel is decadal ($T = 4$: 1981, 1991, 2001, 2011). "Trends" and "lags" are measured in decades, and degrees of freedom in the time dimension are scarce.

B.1. The panel estimator

The estimating equation is the two-way fixed-effects panel of the main text, for a spatial unit i in state r and decade t:

$$Y_{irt} \;=\; \boldsymbol{\beta}'\mathbf{T}_{irt} \;+\; \boldsymbol{\phi}'\mathbf{P}_{irt} \;+\; \mu_i \;+\; \lambda_t \;+\; \gamma_r{\cdot}t \;+\; \varepsilon_{irt}, \tag{A36}$$

where $\mathbf{T}_{irt} = (T_{\min}, T_{\max})'_{irt}$ is the diurnal temperature pair, $\mathbf{P}_{irt}$ the contemporaneous rainfall control, μ_i unit fixed effects, λ_t decade fixed effects, and $\gamma_r{\cdot}t$ a state-specific linear trend. Collect the regressors of interest into $\mathbf{x}_{irt} = (\mathbf{T}'_{irt}, \mathbf{P}'_{irt})'$ and the nuisance terms into $\mathbf{z}_{irt}$. By Frisch–Waugh–Lovell, $\widehat{\boldsymbol{\theta}} = (\widehat{\boldsymbol{\beta}}', \widehat{\boldsymbol{\phi}}')'$ is the least-squares coefficient of the residualized outcome on the residualized regressors,

$$\widehat{\boldsymbol{\theta}} \;=\; \big(\widetilde{\mathbf{X}}'\widetilde{\mathbf{X}}\big)^{-1}\widetilde{\mathbf{X}}'\widetilde{\mathbf{Y}}, \qquad \widetilde{\mathbf{X}} \equiv \mathbf{M}_Z\mathbf{X}, \quad \widetilde{\mathbf{Y}} \equiv \mathbf{M}_Z\mathbf{Y}, \tag{A37}$$

with $\mathbf{M}_Z$ the annihilator of the stacked nuisance design. The point estimate is therefore invariant to how the variance is computed; only the standard error changes, which is what lets us hold the estimate fixed and sweep the inference procedure below.

B.2. Variance estimators

Write $\widehat{\varepsilon} = \widetilde{\mathbf{Y}} - \widetilde{\mathbf{X}}\widehat{\boldsymbol{\theta}}$ and $\mathbf{Q} = \widetilde{\mathbf{X}}'\widetilde{\mathbf{X}}$. All estimators are of the sandwich form $\mathbf{Q}^{-1}\,\widehat{\boldsymbol{\Omega}}\,\mathbf{Q}^{-1}$ and differ only in the meat $\widehat{\boldsymbol{\Omega}}$. *District clustering* groups observations by district g and allows arbitrary

within-district correlation across decades while assuming independence across districts:

$$\widehat{\boldsymbol{\Omega}}_{\text{cl}} \;=\; \sum_{g} \Big(\textstyle\sum_{irt\in g} \widetilde{\mathbf{x}}_{irt}\,\widehat{\varepsilon}_{irt}\Big)\Big(\textstyle\sum_{irt\in g} \widetilde{\mathbf{x}}_{irt}\,\widehat{\varepsilon}_{irt}\Big)'. \tag{A38}$$

Conley spatial–temporal HAC replaces the cluster indicator with a kernel in geographic distance and time. For observations $a = (i,t)$ and $b = (j,s)$ with great-circle distance d_{ij} and decadal separation $|t-s|$,

$$\widehat{\boldsymbol{\Omega}}_{\text{Conley}} \;=\; \sum_{a}\sum_{b} K_{ab}\,\widetilde{\mathbf{x}}_a\widehat{\varepsilon}_a\,\widehat{\varepsilon}_b\widetilde{\mathbf{x}}_b', \qquad K_{ab} = \underbrace{\mathbf{1}\{d_{ij}\le \bar{d}\}}_{\text{spatial}} \cdot \underbrace{\Big(1-\tfrac{|t-s|}{L+1}\Big)_{+}}_{\text{Bartlett, lag } L}, \tag{A39}$$

with spatial cutoff $\bar{d}$ (km) and time-lag cutoff L (decades). With $\bar{d}\to 0$ the estimator reduces to a within-unit Bartlett HAC; with $L \ge T-1$ and $\bar{d}$ small it approaches (A38) with triangular rather than uniform weights on the within-unit lags, which is why, in Table A2, the Conley standard error at 50 km and four lags is marginally smaller than, but otherwise tracks, the district-clustered one. Both are evaluated on the residualized system (A37); inference is normal-based.

B.3. Specification selection by simulated size

Let $\mathcal{C} = \{c\}$ index the candidate procedures—combinations of a fixed-effects/trend structure and a variance estimator ((A38) or (A39) at various $\bar{d}, L$). We select the c whose empirical rejection rate of a true null is closest to the nominal 5%. The DGP imposes the null by *randomization inference* on the regressor. For each replication $s = 1,\dots,S$ ($S = 1{,}000$) we draw a uniform random permutation π_s of the spatial units and construct a placebo climate field by reassigning each unit's *entire* time series to another unit,

$$\widetilde{\mathbf{X}}^{(s)}_{irt} \;=\; \mathbf{X}_{\pi_s(i),\,t}, \tag{A40}$$

moving the $(T_{\min}, T_{\max}, \text{rain})$ block together so their joint structure is preserved. Because π_s is independent of the outcome, the true temperature effect is $\boldsymbol{\beta} = \mathbf{0}$ by construction. For each candidate c we estimate (A36) with the placebo in place of the climate regressors and test $H_0 : \boldsymbol{\beta} = \mathbf{0}$ via the Wald statistic $W_c^{(s)} = \widehat{\boldsymbol{\beta}}^{(s)\prime}[\mathbf{R}\,\widehat{\mathbf{V}}_c^{(s)}\,\mathbf{R}']^{-1}\widehat{\boldsymbol{\beta}}^{(s)}$, where $\mathbf{R}$ selects the temperature coefficients. The empirical size is the rejection frequency $\widehat{\alpha}_c = S^{-1}\sum_s \mathbf{1}\{p(W_c^{(s)}) < 0.05\}$, and the preferred procedure satisfies $\widehat{\alpha}_c \approx 0.05$. Table A2 reports $\widehat{\alpha}_c$ across the Census-town outcomes and three climate regressor sets (the Census-district and ICRISAT panels behave identically). Unadjusted and heteroskedasticity-robust errors reject at three to four times the nominal rate because the placebo is highly persistent; district clustering restores $\widehat{\alpha}_c \approx 0.05$, and adding state linear trends (or state×decade fixed effects) keeps it there, whereas district-specific trends over-absorb on the short panel.

TABLE A2. Placebo Monte-Carlo Size: Rejection Rates by Variance Estimator (Census Towns)

	Seasonal	Cultivators	Ag. labour	Non-ag.	Total workers
No adjustment (homoskedastic)	0.088	0.144	0.126	0.121	0.127
Heteroskedasticity-robust	0.073	0.106	0.098	0.093	0.080
Cluster by district	0.040	0.052	0.052	0.054	0.048
Cluster by district + district linear trends	0.045	0.041	0.044	0.058	0.036
Cluster by district + state linear trends	**0.037**	**0.055**	**0.041**	**0.059**	**0.048**
Cluster by district + state-by-year FE	0.039	0.048	0.047	0.064	0.048
Cluster by district + district-by-year FE	0.053	0.046	0.059	0.060	0.049
Cluster by state	0.083	0.093	0.068	0.077	0.073
Conley spatial HAC (50 km)	0.154	0.192	0.178	0.168	0.154
Conley spatial HAC (100 km)	0.150	0.209	0.182	0.180	0.158
Conley spatial HAC (200 km)	–	–	–	–	–
Conley spatial HAC (50 km, 2 lags)	0.074	0.106	0.106	0.099	0.110
Conley spatial HAC (50 km, 4 lags)	0.043	0.049	0.050	0.051	0.049
Replications			1000		

Notes: Empirical rejection rate $\widehat{\alpha}_c$ of a true null at the nominal 5% level, from 1,000 placebo replications that permute the climate series across spatial units (A40), for each variance estimator (rows) and Census-town outcome (columns). The bold row is the preferred specification—district clustering with state-specific linear trends—which restores nominal size; the Census-district and ICRISAT panels behave identically. "–" marks cells the Conley estimator at that radius does not return.

B.4. What the size simulation does and does not identify

The placebo (A40) relocates whole series across units. It *preserves* each series' marginal distribution, its within-unit serial persistence, and the common secular time profile, but it *eliminates* the spatial correlation of climate, since neighboring units receive unrelated donors. Two consequences follow. First, the spatial radius $\bar{d}$ in (A39) has almost nothing to correct, which is why the rejection rates in Table A2 are nearly flat between 50 and 100 km; the temporal lag L does the work. Second, district clustering and a Conley HAC with enough decadal lags coincide in size because both absorb exactly the within-unit serial dependence that this DGP contains. The simulation cannot rank the two on spatial adequacy, the dimension on which they genuinely differ in the real data, where heat shocks are spatially correlated across nearby districts (Bhattacharjee and Bose 2024) and district clustering, unlike Conley, assumes cross-unit independence. A sharper experiment, left to future work, would permute climate in spatial blocks (region or state) so that the placebo retains spatial correlation. We therefore read the diagnostic as establishing the need to handle serial persistence, which the preferred estimator does, and report the Conley radius sweep of Table A5 as the complementary check on the spatial dimension.

B.5. Why state-specific linear trends

The term $\gamma_r \cdot t$ in (A36) is an economic restriction, not only a variance device. Over 1981–2011 Indian states followed divergent and largely smooth trajectories of structural transformation, driven by state-level policy regimes, infrastructure, and agro-ecological endowments, and these trajectories are correlated with the slow secular warming that the decadal moving-average climate also tracks. Absorbing a state-specific linear trend purges this low-frequency co-movement, so that $\boldsymbol{\beta}$ is identified from deviations of a unit's outcome from its state's own linear path. It is the least aggressive control that accomplishes this: the more saturated alternative, state$\times$decade fixed effects, additionally discards *all* within-state-decade variation—including the spatially correlated climate signal of interest—and so risks absorbing the treatment itself; the fixed-effects robustness in

Tables A11–A12 shows the estimates are nonetheless stable across this choice. The size simulation and this argument are complementary: the state trend is the specification that both restores nominal size against the persistent placebo and carries a substantive interpretation.

B.6. Diagnostics on the diurnal basis

Three exhibits bear on whether the two temperature margins can be separated at all. Table A3 shows that the surface-air pair is near-orthogonal in its estimating variation once the fixed effects and trends are absorbed (within-transformed VIF ≈ 1, condition number 1.1), whereas the wet-bulb pair stays collinear within transformation, which is why its daytime coefficient is the fragile one. Table A4 reparameterizes the baseline into the diurnal mean and range: the range carries the labor reallocation and the harvest-price response, while the common cultivated-area contraction loads on the mean. Table A5 sweeps the Conley radius to 200 km, the residual-correlogram decay distance for districts; the night/day equality test on the agricultural wage-labour share rejects at every radius.

TABLE A3. Multicollinearity Diagnostics: Surface-Air vs Wet-Bulb

	Surface-air (T_{min}, T_{max})		Wet-bulb (T_{wmin}, T_{wmax})	
	Raw	Within	Raw	Within
Correlation ρ	0.78	-0.09	0.97	0.84
Variance-inflation factor	3.3	1.0	21.6	3.5
Condition number	3.3	1.1	9.2	3.4

Notes: Correlation, variance-inflation factor, and condition number for the temperature pair in the district decomposition—surface-air (T_{min}, T_{max}) and wet-bulb (T_{wmin}, T_{wmax})—raw and after the baseline within transformation (district and decade fixed effects, state linear trend, rainfall included). The surface-air baseline is near-orthogonal in its estimating variation (within VIF near 1); the wet-bulb pair stays collinear within transformation, explaining the fragility of its daytime coefficient.

TABLE A4. Diurnal Mean and Range Decomposition

Panel A. ICRISAT Districts (1981-2011)

	Cultivated Area (1)	Production (2)	Harvest Price Index (3)	Field Wage (4)	Yield Index (5)
Diurnal Mean	-0.444***	-0.366***	0.088*	-0.157*	0.006
	(0.0848)	(0.0950)	(0.0453)	(0.0898)	(0.0647)
Diurnal Range	-0.003	-0.038	0.086***	-0.041	-0.049
	(0.0289)	(0.0427)	(0.0236)	(0.0477)	(0.0300)
Observations	1,243	1,243	1,191	1,115	1,211
Dep Var. Mean	5.45	5.77	5.66	2.86	7.14

Panel B. Census Districts (1981-2011)

	Ag. Labour (1)	Cultivator (2)	Seasonal (3)	Non-Agriculture (4)	Log Workers (5)
Diurnal Mean	0.018	-0.038**	-0.001	0.021	0.159
	(0.0128)	(0.0169)	(0.0178)	(0.0171)	(0.1162)
Diurnal Range	-0.035***	0.009	0.032***	-0.006	-0.096
	(0.0070)	(0.0092)	(0.0084)	(0.0085)	(0.0625)
Observations	1,187	1,187	1,187	1,187	1,187
Dep Var. Mean	0.18	0.32	0.17	0.32	13.45
Rainfall Control	Y	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y	Y

Notes: Reparameterizes the baseline (T_{min}, T_{max}) decomposition into the growing-season diurnal mean $\frac{1}{2}(T_{min} + T_{max})$ and range $(T_{max} - T_{min})$. The transformation is exact and linear, so the range coefficient equals $\frac{1}{2}(\beta_{T_{max}} - \beta_{T_{min}})$ and its significance is identical to the night/day equality test of the main decomposition. The diurnal range carries the labor reallocation and the harvest-price response; the common cultivated-area contraction loads on the mean. Decadal panel, unit and decade fixed effects and a state $\times$ decade linear trend, precipitation control; district-clustered standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A5. Conley Spatial-HAC Standard Errors: Radius Sweep

	Coefficient	Standard error by estimator				
		Cluster	50 km	100 km	150 km	200 km
Panel A. ICRISAT Districts						
Cultivated area						
T_{min}	-0.218***	(0.0459)	(0.0459)	(0.0524)	(0.0578)	(0.0611)
T_{max}	-0.225***	(0.0562)	(0.0560)	(0.0630)	(0.0684)	(0.0689)
Harvest price index						
T_{min}	-0.042	(0.0311)	(0.0311)	(0.0359)	(0.0396)	(0.0446)
T_{max}	0.130***	(0.0341)	(0.0345)	(0.0361)	(0.0392)	(0.0430)
Field wage						
T_{min}	-0.038	(0.0603)	(0.0599)	(0.0689)	(0.0792)	(0.0857)
T_{max}	-0.119*	(0.0703)	(0.0677)	(0.0795)	(0.0898)	(0.0971)
Panel B. Census Districts						
Ag. wage labour share						
T_{min}	0.044***	(0.0096)	(0.0095)	(0.0106)	(0.0115)	(0.0129)
T_{max}	-0.026***	(0.0094)	(0.0094)	(0.0115)	(0.0127)	(0.0141)
Seasonal share						
T_{min}	-0.033***	(0.0115)	(0.0115)	(0.0141)	(0.0165)	(0.0187)
T_{max}	0.032**	(0.0129)	(0.0130)	(0.0150)	(0.0173)	(0.0178)
Non-agriculture share						
T_{min}	0.017	(0.0114)	(0.0112)	(0.0114)	(0.0111)	(0.0102)
T_{max}	0.004	(0.0127)	(0.0125)	(0.0131)	(0.0129)	(0.0129)

Notes: Baseline decadal-panel point estimates; the standard-error columns sweep the Conley spatial–temporal HAC radius from 50 to 200 km (four decadal time lags throughout), spanning the residual spatial-correlation decay distance for districts. Standard errors rise with the radius. The night/day equality test on the agricultural wage-labour share rejects at every radius ($F = 25.2$, 19.8, 15.4, and 12.4 at 50, 100, 150, and 200 km, all $p < 0.001$); the nighttime labour and daytime harvest-price coefficients remain significant at 5% throughout, while the daytime labour and both seasonal coefficients thin to the 10% level at the widest (200 km) radius. Coefficient stars are district-clustered. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix C. Data Sources and Construction

I harmonize four sources onto a fixed spatial frame: the India Meteorological Department (IMD) gridded climate, the ICRISAT District Level Database, the Indian Census (district and town tables), and the NSS Employment and Unemployment schedules. The unit of analysis is the 2011 Census district (and, for the town exhibits, the Census town). This section documents the concordance, the spatial assignment of climate, and the construction of each sample, and closes with the summary statistics and the coefficient tables behind the main-text figures. The replication package documents the file-level lineage from raw sources through the build and estimation scripts.

C.1. District concordance

India's district boundaries change across rounds through splits and, less often, mergers, so I map every Census round (1961–2011) onto a fixed 2011 frame using a name-and-code crosswalk (the `indiabridge` utility, keyed on the 1981, 2001, and 2011 district codes). Tables for 1961–1991 come from the Vanneman compilation and for 2001–2011 from the digitized Primary Census Abstract. The concordance is a least-common-unit (inner) join: a district enters the panel only if it matches cleanly across all sources. I do not apportion population between split parents and children; instead, historical districts that cannot be uniquely assigned to a single 2011 unit (for example composite Vanneman records spanning more than one successor) are dropped rather than divided, trading coverage for boundary stability. Eight states are excluded for want of a usable 1981 identifier (Arunachal Pradesh, Goa–Daman–Diu, Manipur, Meghalaya, Nagaland, Tripura, Sikkim, and Mizoram). A handful of district-decades are missing by design and coded as such rather than zero: Assam in 1981 and Jammu & Kashmir in 1991 (Census not conducted), Calcutta in 1961–1991, and the Andaman & Nicobar Islands in 1961. The harmonized panel contains 300 districts; the decadal panel sample is 1,187 district-decades and the long-difference sample is 293 districts. Worker categories are reconciled across the 1981/1991 and 2001/2011 definitions: total workers are main plus marginal workers, agricultural workers are cultivators plus agricultural labourers, and non-agricultural workers are the residual of main workers. All shares are winsorized at the 1st and 99th percentiles within decade.

A known enumeration change affects the marginal-worker category—the "seasonal" margin in the main text. The 2001 Census captured marginal and casual work far more completely than 1991, so the marginal-worker share roughly doubles across the 1991–2001 boundary (rural district seasonal shares rise from about 0.12 to 0.26, and balanced-town shares from about 0.03 to 0.12; Table A7). Two features contain this break. First, the *headline* divergence is identified off the *main*-worker categories—agricultural wage labor and self-cultivation—which are defined consistently throughout and are not reclassified by the marginal-worker change; only the seasonal-sink channel draws on the marginal count, and there the night/day split is the weaker of the reported results. Second, the common component of the 2001 shift is a level break shared across districts within a round, which the decade fixed effects in the panel absorb; the long difference, which straddles the 1991–2001 boundary, is the more exposed design, and its estimates are reported alongside the panel throughout so the two can be read against one another. The per-capita denomination of §D.1 is immune to the break and reproduces the results. Worker shares are therefore best read as main-worker reallocations, with the seasonal margin as the flexible—and more definition-sensitive—residual.

C.2. Climate exposure and IMD assignment

Daily temperature and precipitation come from the IMD gridded products (air temperature at 1°, precipitation at 0.25°, 1951–2023). I assign grid points to each district or town centroid by inverse-distance-squared weighting: for every centroid I take all grid points within 200 km (100 km for towns), weight each by $1/d^2$ normalized to sum to one, and form the weighted average. The cutoffs match the distance at which the residual spatial correlation of the outcomes decays (about 150–200 km for districts and 100 km for towns). Because the weights decay with the square of distance, points beyond roughly 100 km contribute negligibly, so the assignment is effectively local and insensitive to the outer cutoff—consistent with the residual correlograms reported in the main text, whose spatial correlation decays within about 100 km. The harmonized grid comprises 362 points. Following Liu, Shamdasani, and Taraz (2023), decadal exposure is the mean over the growing seasons of the preceding ten years, so that climate and outcomes share a common ten-year horizon at each Census. The growing season spans June–February (the kharif and rabi cropping cycles); months are tagged kharif (June–October), rabi (November–February), and lean (March–May), and the lean months are excluded from the growing-season aggregates. The baseline margins are the growing-season daily minimum and maximum surface-air temperatures, T_{min} and T_{max}, entered simultaneously. As a scalar alternative I construct growing-season degree days, accumulating daily mean temperature between 8 and 32 °C (Blakeslee and Fishman 2015; Schlenker, Hanemann, and C. 2006), which average over the diurnal cycle and so collapse the night and day margins into one. Standard errors are clustered at the district level throughout; the main text also reports spatially-corrected (Conley 1999) standard errors as a check on residual spatial autocorrelation.

C.3. Wet-bulb temperatures

The humidity-adjusted measures use *no external humidity series*: relative humidity is recovered from the daily temperature pair itself and converted to a wet-bulb temperature by the empirical formula of Stull (2011). Daily actual and saturation vapour pressures are computed from the Tetens equation, $e(T) = 6.1078 \exp\big[17.269\,T/(237.3+T)\big]$, using the daily minimum as a proxy for the dewpoint: $e_a = e(T_{min})$ and $e_s = e(T_{max})$, so daytime relative humidity is $\mathrm{RH} = 100\,e_a/e_s$, capped at 100%. The daytime wet-bulb temperature then follows Stull (2011),

$$T_{wmax} = T_{max}\arctan\big[0.151977\,(\mathrm{RH}+8.313659)^{1/2}\big] + \arctan(T_{max}+\mathrm{RH}) - \arctan(\mathrm{RH}-1.676331) + 0.00391838\,\mathrm{RH}^{3/2}\arctan(0.023101\,\mathrm{RH}) - 4.686035,$$

while $T_{wmin} = T_{min}$ (at the daily minimum the dewpoint proxy implies RH $\approx$ 100%, where wet-bulb and dry-bulb temperatures coincide). The wet-bulb minimum and maximum are aggregated to growing-season decadal means exactly as the surface-air measures. Because T_{wmin} and T_{wmax} are deterministic functions of T_{min} and T_{max}, the wet-bulb specifications are a nonlinear robustness re-expression of the surface-air pair rather than an independent humidity measurement, and are reported as a supplementary check.

C.4. Agricultural data: ICRISAT

Cultivated area, production, yields, and harvest prices come from the ICRISAT District Level Database (annual, 1966–present), averaged over the same ten-year windows preceding each Census. ICRISAT's agro-economic districts are not nested in Census boundaries, so I match them to 2011 Census districts in stages, by decreasing confidence: a set of manual overrides for known

splits and renamings, then exact normalized-name matches, then substring matches, then fuzzy (edit-distance) matches at a 0.6 similarity threshold; Telangana districts are folded into Andhra Pradesh to match the pre-2014 frame. Of 313 ICRISAT districts, 237 match exactly, 18 by substring, 33 by fuzzy rule, and 25 by manual override, with 0 unmatched. The estimation panel covers 311 districts. Crop outcomes are built for rice, wheat, maize, sorghum, barley, and millets, and for a pooled cereal index.

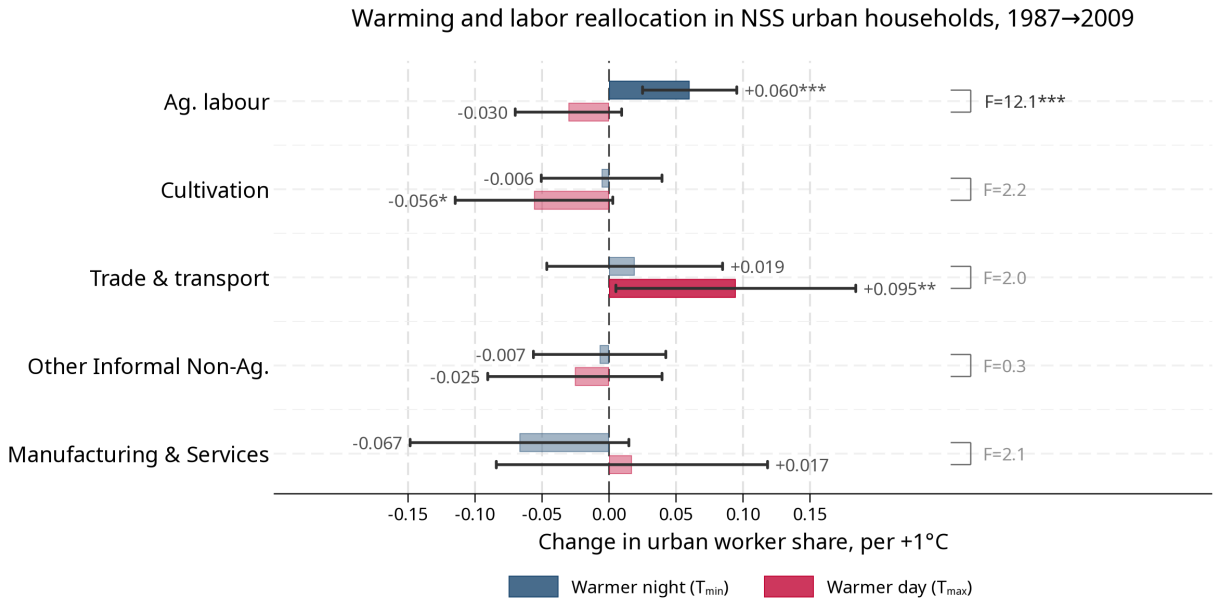
C.5. Town panel

The town data are digitized from the Primary Census Abstracts, 1961–2011. A Census town must satisfy the standard criteria (population at least 4,000, at least 75% of male workers outside agriculture, and density at least 400 persons per km^2). Towns are not a stable universe: places enter as large villages "graduate" to town status and, occasionally, exit. I therefore use two samples. The town *panel* uses the full set of towns observed from 1981 onward (8,786 towns, 16,215 town-decades); the town *long-difference* restricts to the balanced set present in every round since 1981 (2,969 towns), so the endpoint comparison reflects intensive-margin adjustment within continuing towns rather than the reclassification of new ones. Because the appearance of new towns is itself partly an outcome of growth, the balanced long-difference is the more conservative design for this composition concern, and the panel and balanced results are reported alongside one another. Occupations use the same four-way split as the district data (cultivators, agricultural labourers, non-agricultural workers, and marginal workers).

C.6. NSS employment surveys (IPUMS)

To resolve the destination sector of displaced labour—which the Census's 1981 occupation definitions do not—I use IPUMS-International extracts of the NSS Employment and Unemployment schedules (Ruggles et al. 2024), for rounds spanning 1987, 1999, and 2009. The unit is the harmonized sub-national region (`geo2`), restricted to urban households as a benchmark to the Census town sample (416 regions, 1,237 region-rounds). These rounds are *not* merged onto Census years: each round is treated as its own observation and paired with climate averaged over its own preceding ten-year window (for example 1977–1986 for the 1987 round), so the worker-share levels are never assumed to proxy a specific Census year. Occupations are grouped into five mutually exclusive sectors that partition the urban workforce: agricultural wage labour, self-cultivation, trade and transport, other informal work (construction, mining, utilities, hospitality, and personal services), and formal manufacturing and services. NSS results appear only as suggestive evidence. Figure A1 visualises the sectoral destination; Table A6 reports the full coefficients.

FIGURE A1. Where heat-displaced urban labour goes (NSS/IPUMS)



Notes: Suggestive evidence for future work. Panel response of each urban worker share to a 1°C rise in the growing-season nighttime minimum (T_{min}, navy) and daytime maximum (T_{max}, cranberry), from three rounds of the NSS Employment–Unemployment schedules (IPUMS-International, 1987–2009), restricted to urban households. The five shares partition the urban workforce. Spikes are 95% confidence intervals; full-colour bars are significant at 5%. The brace on each sector reports the night-versus-day equality test $F(T_{min} = T_{max})$, significant only for agricultural wage labour. Full coefficients are in Table A6.

TABLE A6. NSS Urban Households: Sectoral Destination of Displaced Labour

	Agriculture		Informal Non-Agriculture		Formal
	Ag. Labour (1)	Cultivation (2)	Trade, Transport & Comm. (3)	Other Informal (4)	Manuf. & Services (5)
T_{min}	0.060***	-0.006	0.019	-0.007	-0.067
	(0.0179)	(0.0230)	(0.0335)	(0.0252)	(0.0416)
	[0.0169]	[0.0222]	[0.0314]	[0.0242]	[0.0409]
T_{max}	-0.030	-0.056*	0.095**	-0.025	0.017
	(0.0203)	(0.0300)	(0.0457)	(0.0332)	(0.0516)
	[0.0194]	[0.0288]	[0.0436]	[0.0321]	[0.0507]
Observations	1,163	1,163	1,163	1,163	1,163
Dep Var. Mean	0.04	0.07	0.31	0.14	0.44
F-stat ($T_{min} = T_{max}$)	12.08	2.17	2.03	0.26	2.07
Rainfall Control	Y	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y	Y

Notes: Urban worker shares in the NSS Employment–Unemployment rounds (IPUMS-International, 1987–2009), restricted to urban households and classified into five mutually exclusive sectors that partition the urban workforce: agricultural wage labour and self-cultivation (cultivators and unpaid family workers); two informal non-agricultural sectors—trade/transport/communication, and other informal work (construction, mining, utilities, hospitality, and personal and household services)—grouped as urban casual labour; and formal manufacturing and high-skill services. The NSS urban frame differs from the Census town definition—it retains the peri-urban self-cultivation a Census town (by construction over 75% non-agricultural) excludes—so the two urban measures agree on the agricultural wage-labour margin but not on self-cultivation. Decadal panel with district and decade fixed effects and a state $\times$ decade linear trend, and precipitation controls. Parentheses report district-clustered standard errors; brackets report Conley spatial–temporal HAC standard errors (50 km, four decadal lags). The F-statistic tests $T_{min} = T_{max}$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.7. Rainfall control

Every specification controls for precipitation. Rainfall is taken from the same IMD gridded daily product (0.25°) and assigned to districts and towns with the identical inverse-distance-squared scheme, then aggregated to growing-season totals and averaged over the same preceding-decade window as temperature. No alternative gridded product (APHRODITE, CRU) is used. Because rainfall covaries with both temperature margins, controlling for it over the matched window is what isolates the partial temperature effects.

C.8. Summary statistics

Table A7 reports means and standard deviations of the main outcomes and of the temperature and rainfall regressors, separately for the decadal panel and the long-difference samples, with district and observation counts for each.

TABLE A7. Summary Statistics: District and Town, Panel and Long-Difference Samples

	Decadal panel (levels) 1981–2011		Long difference (Δ 2001–11 vs. 1981–91)	
	District	Town	District	Town
Cultivators (share)	0.323	0.080	-0.139	-0.058
	(0.158)	(0.091)	(0.080)	(0.057)
Agricultural labourers (share)	0.180	0.088	-0.049	-0.053
	(0.111)	(0.097)	(0.057)	(0.060)
Seasonal / marginal (share)	0.173	0.085	0.139	0.102
	(0.111)	(0.079)	(0.086)	(0.065)
Non-agricultural (share)	0.324	0.745	0.048	0.010
	(0.176)	(0.178)	(0.070)	(0.089)
Nighttime T_{min} (°C)	17.95	18.23	0.28	0.29
	(2.39)	(2.22)	(0.24)	(0.24)
Daytime T_{max} (°C)	29.71	29.89	0.15	0.11
	(1.95)	(1.63)	(0.19)	(0.20)
Rainfall (growing-season mean)	1.0	1.0	0.0	0.0
	(0.5)	(0.5)	(0.1)	(0.1)
Units (districts / towns)	300	2,969	299	2,969
Observations	1,187	11,876	299	2,969

Notes: Means with standard deviations in parentheses, for the decadal panel (levels, 1981–2011) and the long-difference cross-section (Δ 2001–11 vs. 1981–91), each split into the district (total-population) and Census-town samples. Worker shares are of total workers in the unit; temperature and rainfall are growing-season ten-year moving averages. The town columns use the balanced panel of towns present in every round since 1981—the long-difference sample—so the panel-levels and long-difference town statistics describe the same 2,969 towns. The district panel counts one more unit than the long difference (300 vs. 299): one district is observed only at the endline and so enters the panel but cannot be differenced. These are descriptive samples; the estimation samples are marginally smaller after dropping observations with missing climate covariates.

C.9. Coefficients behind the main-text figures

Figures 2 and 3 of the main text plot the long-difference agricultural wage-labour coefficients for the rural and urban populations of each district and for Census towns. Tables A8 and A9 report the full estimates behind them, in the surface-air baseline, for all four worker shares and log total workers. The district table decomposes each district's workforce into its total, rural, and urban populations (the within-district residence split).

TABLE A8. District Worker Reallocation, Long Difference (1981–2011), by Residence

	Share of Workers				
	Ag. Labour	Cultivator	Seasonal	Non-Agriculture	Log Total Workers
	(1)	(2)	(3)	(4)	(5)
Panel A. Total population					
ΔT_{min}	**0.075**	-0.024	**-0.074**	0.026	0.297
	(3.58)	(-0.79)	(-2.74)	(0.94)	(1.24)
ΔT_{max}	**-0.062**	-0.008	0.049	0.020	0.041
	(-2.51)	(-0.22)	(1.47)	(0.55)	(0.16)
F ($\Delta T_{min} = \Delta T_{max}$)	26.07	0.16	12.34	0.03	0.83
p-value	<0.001	0.687	<0.001	0.865	0.363
Observations	293	293	293	293	293
Panel B. Rural population					
ΔT_{min}	**0.096**	-0.035	**-0.064**	0.007	0.211
	(4.12)	(-1.11)	(-2.25)	(0.37)	(0.91)
ΔT_{max}	**-0.068**	0.011	0.056	-0.001	0.025
	(-2.52)	(0.28)	(1.58)	(-0.02)	(0.10)
F ($\Delta T_{min} = \Delta T_{max}$)	30.87	1.24	11.36	0.07	0.48
p-value	<0.001	0.266	<0.001	0.794	0.489
Observations	291	291	291	291	293
Panel C. Urban population					
ΔT_{min}	**0.034**	-0.013	**-0.028**	0.007	0.632
	(2.64)	(-0.89)	(-2.09)	(0.25)	(1.75)
ΔT_{max}	0.006	0.015	0.006	-0.028	0.305
	(0.32)	(0.89)	(0.37)	(-0.75)	(0.80)
F ($\Delta T_{min} = \Delta T_{max}$)	1.89	1.69	3.23	0.69	0.60
p-value	0.171	0.194	0.073	0.407	0.439
Observations	291	291	291	291	293
Rainfall Control	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y

Notes: Long-difference estimates of nighttime (ΔT_{min}) and daytime (ΔT_{max}) growing-season surface-air warming on district worker shares, decomposed into the total, rural, and urban populations of each district (the within-district residence split). These are the coefficients behind Figure 2 of the main text: the agricultural wage-labour column of Panels B and C is plotted there. Point estimates with $|t| > 2$ in bold over district-clustered t-statistics; the F-statistic and its p-value test $\Delta T_{min} = \Delta T_{max}$. All specifications include state fixed effects and a precipitation control.

TABLE A9. Town Worker Reallocation, Long Difference (1981–2011)

	Share of Workers				
	Ag. Labour	Cultivator	Seasonal	Non-Agriculture	Log Total Workers
	(1)	(2)	(3)	(4)	(5)
ΔT_{min}	**0.063**	-0.009	-0.016	**-0.038**	0.272
	(5.46)	(-0.86)	(-1.35)	(-2.16)	(1.61)
ΔT_{max}	**0.041**	0.022	0.017	**-0.078**	0.241
	(2.59)	(1.49)	(1.15)	(-3.47)	(1.41)
Observations	2,969	2,969	2,969	2,969	2,969
F-stat ($\Delta T_{min} = \Delta T_{max}$)	1.61	3.38	4.06	2.78	0.11
p-value	0.205	0.067	0.044	0.096	0.738
Rainfall Control	Y	Y	Y	Y	Y
FE / Trend	Y	Y	Y	Y	Y

Notes: Census-town worker reallocation under nighttime (ΔT_{min}) and daytime (ΔT_{max}) growing-season surface-air warming, across the balanced panel of Census towns—the coefficients behind Figure 3 of the main text. The four mutually exclusive worker categories (agricultural wage labour, cultivators, seasonal, non-agricultural) sum to one, so the share coefficients sum to approximately zero; column 5, set apart, reports the long-run change in log total town workers. Point estimates with $|t| > 2$ in bold over district-clustered t-statistics; the F-statistic and its p-value test $\Delta T_{min} = \Delta T_{max}$. State fixed effects and a precipitation control throughout.

Appendix D. Robustness

This section collects the robustness checks the main text cites, one subsection per concern: the outcome denominator and the 1991–2001 enumeration break (§D.1); the fixed-effects and trend structure (§D.2); the climate measure, including wet-bulb temperatures, degree days, and individual versus simultaneous entry of the two margins (§D.3); the crop composition of the cereal index (§D.4); heterogeneity by baseline climate and irrigation (§D.5); and the stability of the town result across the town sample (§D.6). The generated tables re-estimate the district labour (Table 2) and ICRISAT agricultural (Table 1) specifications of the main text in wet-bulb temperatures unless a block notes otherwise.

D.1. Outcome denominator: per-capita worker rates

The worker shares in the main text are denominated by the local workforce: each share is a category's worker count divided by total workers, and the four mutually exclusive categories sum to one. Because total workers include the marginal count, the 1991–2001 enumeration change (§C.1) moves every share's denominator; the population denominator used here is immune to it. Table A10 re-estimates the long difference with outcomes denominated instead by *population*—per-capita worker rates (a category's count divided by population), which sum to the work-participation rate (column 5) rather than to one. The district agricultural wage-labour divergence is unchanged: warmer nights raise, and warmer days lower, the per-capita ag-labour rate—sharply for the rural population ($F = 33$) and faintly for the urban ($F = 2$), the same amplitude gradient as the worker-share results in Figure 2. The per-capita lens also surfaces a participation margin the share cannot: warmer nights raise the rural work-participation rate (column 5, $+0.033$, $F = 7.4$ against a flat daytime coefficient), so the nighttime pull into agricultural wage-labour reflects both a recomposition of the existing workforce and net entry into it. In Census towns the agricultural wage-labour result likewise replicates; the one margin that attenuates under the population denominator is the non-agricultural share, consistent with part of the town worker-share contraction operating through rising agricultural participation rather than a fall in non-agricultural employment per capita.

TABLE A10. Per-Capita Worker Rates, Long Difference (1981–2011)

	Per-capita worker rate (share of population)				
	Ag. Labour	Cultivator	Seasonal	Non-Agriculture	Work Particip.
	(1)	(2)	(3)	(4)	(5)
Panel A. Districts — Total population					
ΔT_{min}	**0.028**	0.013	**-0.039**	0.016	0.020
	(3.20)	(0.95)	(-3.12)	(1.60)	(1.50)
ΔT_{max}	**-0.027**	-0.013	0.016	0.005	-0.016
	(-2.67)	(-0.84)	(1.03)	(0.35)	(-0.99)
F ($\Delta T_{min} = \Delta T_{max}$)	25.43	2.36	11.44	0.59	3.65
p-value	<0.001	0.126	<0.001	0.444	0.057
Observations			293		
Panel B. Districts — Rural population					
ΔT_{min}	**0.039**	0.020	**-0.032**	0.007	**0.033**
	(3.83)	(1.30)	(-2.34)	(0.93)	(2.31)
ΔT_{max}	**-0.032**	-0.009	0.020	-0.004	-0.020
	(-2.84)	(-0.53)	(1.20)	(-0.30)	(-1.19)
F ($\Delta T_{min} = \Delta T_{max}$)	33.12	2.43	9.97	0.67	7.40
p-value	<0.001	0.120	0.002	0.415	0.007
Observations			291		
Panel C. Districts — Urban population					
ΔT_{min}	**0.010**	-0.003	**-0.011**	0.001	-0.002
	(2.55)	(-0.68)	(-2.60)	(0.08)	(-0.33)
ΔT_{max}	0.000	0.002	0.004	-0.006	0.000
	(0.03)	(0.50)	(0.74)	(-0.51)	(0.01)
F ($\Delta T_{min} = \Delta T_{max}$)	2.39	0.75	6.07	0.28	0.06
p-value	0.123	0.387	0.014	0.596	0.810
Observations			291		
Panel D. Census towns					
ΔT_{min}	**0.021**	0.000	-0.004	0.000	**0.019**
	(5.67)	(0.01)	(-0.92)	(0.05)	(2.97)
ΔT_{max}	**0.014**	0.006	0.008	-0.014	0.012
	(2.65)	(1.39)	(1.66)	(-1.50)	(1.13)
F ($\Delta T_{min} = \Delta T_{max}$)	1.55	1.45	4.95	2.30	0.42
p-value	0.213	0.228	0.026	0.130	0.516
Observations			2,969		
Rainfall control	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y

Notes: Long-difference estimates (1981–2011) of growing-season nighttime (ΔT_{min}) and daytime (ΔT_{max}) warming on per-capita worker rates—each category's worker count divided by population. The four mutually exclusive categories (columns 1–4) sum to the work-participation rate (column 5). Panels A–C split each district's population into total, rural, and urban (the within-district residence split); Panel D is the balanced panel of Census towns. All specifications enter T_{min} and T_{max} jointly with a rainfall control and state fixed effects; parentheses report t-statistics from district-clustered standard errors, and coefficients with $|t| > 2$ are in bold. The F-statistic and its p-value test $\Delta T_{min} = \Delta T_{max}$. Compare with the worker-share results in Tables A8 and A9.

D.2. Fixed-effects and trend structure

Tables A11 and A12 vary the absorbed fixed effects and trends around the baseline (unit and decade effects with a state linear trend; §B.5): two-way effects alone, district-specific trends, and state×decade effects. The comparison that matters for §B.5 is between the baseline and the more saturated state×decade effects, and there the estimates are stable: the night/day divergence on the agricultural wage-labour share ($F = 16.3$ against 19.2) and the daytime area contraction survive intact. Dropping the state trend altogether (two-way effects) leaves the low-frequency state trajectories in the error and changes the labour coefficients materially, while district-specific trends, which over-absorb on a four-decade panel (§B.3), drain the labour divergence to insignificance ($F = 1.6$) even as the daytime harvest-price response survives them ($F = 17.6$).

TABLE A11. Census Districts, Panel: Fixed-Effects Structure

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
Unit + year + state trend (baseline)				
T_{wmin}	-0.083***	0.086***	-0.013	0.011
	(0.0216)	(0.0173)	(0.0237)	(0.0221)
	[0.0214]	[0.0171]	[0.0235]	[0.0217]
T_{wmax}	0.095***	-0.079***	-0.028	0.009
	(0.0349)	(0.0253)	(0.0348)	(0.0355)
	[0.0352]	[0.0252]	[0.0344]	[0.0351]
F-stat ($T_{wmin} = T_{wmax}$)	10.68	16.27	0.08	0.00
Observations	1,187	1,187	1,187	1,187
Unit + year (two-way)				
T_{wmin}	0.126***	0.043***	-0.165***	-0.006
	(0.0210)	(0.0157)	(0.0226)	(0.0160)
	[0.0216]	[0.0162]	[0.0234]	[0.0158]
T_{wmax}	-0.310***	0.014	0.274***	0.027
	(0.0343)	(0.0251)	(0.0366)	(0.0272)
	[0.0359]	[0.0263]	[0.0388]	[0.0273]
F-stat ($T_{wmin} = T_{wmax}$)	70.34	0.57	59.65	0.67
Observations	1,187	1,187	1,187	1,187
Unit + year + district trend				
T_{wmin}	-0.012	0.018	0.005	-0.010
	(0.0266)	(0.0188)	(0.0205)	(0.0168)
	[0.0228]	[0.0163]	[0.0176]	[0.0145]
T_{wmax}	0.067*	-0.011	-0.071**	0.014
	(0.0389)	(0.0276)	(0.0295)	(0.0280)
	[0.0332]	[0.0238]	[0.0251]	[0.0241]
F-stat ($T_{wmin} = T_{wmax}$)	1.63	0.42	2.52	0.33
Observations	1,187	1,187	1,187	1,187
Unit + state×year				
T_{wmin}	-0.100***	0.115***	-0.014	0.002
	(0.0291)	(0.0226)	(0.0327)	(0.0294)
	[0.0284]	[0.0220]	[0.0316]	[0.0284]
T_{wmax}	0.093*	-0.130***	0.002	0.031
	(0.0479)	(0.0354)	(0.0514)	(0.0479)
	[0.0476]	[0.0347]	[0.0499]	[0.0462]
F-stat ($T_{wmin} = T_{wmax}$)	6.76	19.15	0.04	0.15
Observations	1,164	1,164	1,164	1,164
Dep. var. mean	0.173	0.180	0.323	0.324

Notes: Blocks vary the absorbed fixed effects and trends: unit+year+state trend (baseline), two-way unit+year, unit+year+district trend, and unit+state×year. Outcomes are shares of total district workers. Decadal panel (1981–2011) with unit and decade fixed effects and a state × decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.2). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE A12. ICRISAT Districts, Panel: Fixed-Effects Structure

	Cultivated Area (1)	Yield Index (2)	Field Wage (3)	Price Index (4)
Unit + year + state trend (baseline)				
T_{wmin}	0.090	0.153**	0.137	-0.237***
	(0.0781)	(0.0773)	(0.1243)	(0.0598)
	[0.0778]	[0.0770]	[0.1204]	[0.0598]
T_{wmax}	-0.579***	-0.194	-0.328*	0.366***
	(0.1534)	(0.1280)	(0.1928)	(0.0936)
	[0.1529]	[0.1270]	[0.1854]	[0.0946]
F-stat ($T_{wmin} = T_{wmax}$)	9.01	3.07	2.28	16.59
Observations	1,243	1,211	1,115	1,191
Unit + year (two-way)				
T_{wmin}	0.709***	0.415***	-0.088	0.130
	(0.0943)	(0.0551)	(0.0803)	(0.0799)
	[0.0996]	[0.0568]	[0.0756]	[0.0797]
T_{wmax}	-1.239***	-0.523***	0.058	-0.081
	(0.1703)	(0.1046)	(0.1399)	(0.1420)
	[0.1788]	[0.1076]	[0.1426]	[0.1427]
F-stat ($T_{wmin} = T_{wmax}$)	56.99	37.55	0.48	0.95
Observations	1,243	1,211	1,115	1,191
Unit + year + district trend				
T_{wmin}	0.044	-0.032	0.169	-0.288***
	(0.0742)	(0.0984)	(0.2244)	(0.0726)
	[0.0651]	[0.0862]	[0.1716]	[0.0626]
T_{wmax}	-0.250**	0.003	-0.359	0.449***
	(0.1142)	(0.1409)	(0.2995)	(0.1117)
	[0.1031]	[0.1234]	[0.2322]	[0.0964]
F-stat ($T_{wmin} = T_{wmax}$)	2.78	0.02	1.07	17.58
Observations	1,243	1,211	1,115	1,191
Unit + state×year				
T_{wmin}	0.166	0.273***	0.063	-0.086
	(0.1143)	(0.1012)	(0.1354)	(0.0706)
	[0.1112]	[0.0973]	[0.1242]	[0.0690]
T_{wmax}	-0.806***	-0.313	-0.415*	0.088
	(0.2418)	(0.1914)	(0.2136)	(0.1136)
	[0.2343]	[0.1845]	[0.1969]	[0.1115]
F-stat ($T_{wmin} = T_{wmax}$)	7.86	4.26	2.02	0.95
Observations	1,243	1,211	1,114	1,190
Dep. var. mean	5.451	7.142	2.856	5.660

Notes: Blocks vary the absorbed fixed effects and trends: unit+year+state trend (baseline), two-way unit+year, unit+year+district trend, and unit+state×year. Outcomes are log levels for the cereal index (rice, wheat, barley, maize, millets, sorghum). Decadal panel (1981–2011) with unit and decade fixed effects and a state × decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.2). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.3. Alternative climate measures

Tables A13 and A14 re-estimate the two main-text panels replacing the climate regressors: the wet-bulb pair (T_{wmin}, T_{wmax}), the surface-air pair of the main text, and growing-season and calendar-year degree days (8–32 °C). A single degree-days measure, which averages over the diurnal cycle, collapses the night/day opposition into one coefficient that takes the sign of the nighttime margin on the labour shares and of the common contraction on cultivated area. The wet-bulb pair reproduces the sign pattern of the surface-air baseline with roughly twice the labour coefficients and, on the ICRISAT side, raises cereal yields outright under nighttime warming (+0.15) with a nighttime price decline—the land-augmenting signature discussed in Remark A10—whereas the surface-air night margin is on net contractionary. Table A15 enters the two margins one at a time: because the wet-bulb pair is highly correlated ($\rho = 0.93$; Table A3), individual entry loads each margin's coefficient with the omitted other, and the simultaneous specification is what separates them.

TABLE A13. Census Districts, Panel: Alternative Climate Measures

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
Wet-bulb T_{wmin}/T_{wmax}, growing season (baseline)				
T_{wmin}	-0.083***	0.086***	-0.013	0.011
	(0.0216)	(0.0173)	(0.0237)	(0.0221)
	[0.0214]	[0.0171]	[0.0235]	[0.0217]
T_{wmax}	0.095***	-0.079***	-0.028	0.009
	(0.0349)	(0.0253)	(0.0348)	(0.0355)
	[0.0352]	[0.0252]	[0.0344]	[0.0351]
F-stat ($T_{wmin} = T_{wmax}$)	10.68	16.27	0.08	0.00
Observations	1,187	1,187	1,187	1,187
Dry-bulb T_{min}/T_{max}, growing season				
T_{min}	-0.033***	0.044***	-0.028**	0.017
	(0.0115)	(0.0096)	(0.0125)	(0.0114)
	[0.0115]	[0.0095]	[0.0123]	[0.0112]
T_{max}	0.032**	-0.026***	-0.010	0.004
	(0.0129)	(0.0094)	(0.0126)	(0.0127)
	[0.0130]	[0.0094]	[0.0125]	[0.0125]
F-stat ($T_{min} = T_{max}$)	14.66	24.91	0.85	0.54
Observations	1,187	1,187	1,187	1,187
Degree days, growing season				
Degree Days	-0.020	0.036***	-0.041**	0.025
	(0.0182)	(0.0138)	(0.0178)	(0.0178)
	[0.0186]	[0.0136]	[0.0176]	[0.0175]
Observations	1,187	1,187	1,187	1,187
Degree days, calendar year				
Degree Days	-0.011	0.033***	-0.050***	0.026*
	(0.0151)	(0.0117)	(0.0148)	(0.0151)
	[0.0154]	[0.0115]	[0.0147]	[0.0149]
Observations	1,187	1,187	1,187	1,187
Dep. var. mean	0.173	0.180	0.323	0.324

Notes: Each block re-estimates the district worker-share decadal-panel specification (Table 2 of the main text) replacing the climate regressors. Outcomes are shares of total district workers. Decadal panel (1981–2011) with unit and decade fixed effects and a state × decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.2). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE A14. ICRISAT Districts, Panel: Alternative Climate Measures

	Cultivated Area (1)	Yield Index (2)	Field Wage (3)	Price Index (4)
Wet-bulb T_{wmin}/T_{wmax}, growing season (baseline)				
T_{wmin}	0.090	0.153**	0.137	-0.237***
	(0.0781)	(0.0773)	(0.1243)	(0.0598)
	[0.0778]	[0.0770]	[0.1204]	[0.0598]
T_{wmax}	-0.579***	-0.194	-0.328*	0.366***
	(0.1534)	(0.1280)	(0.1928)	(0.0936)
	[0.1529]	[0.1270]	[0.1854]	[0.0946]
F-stat ($T_{wmin} = T_{wmax}$)	9.01	3.07	2.28	16.59
Observations	1,243	1,211	1,115	1,191
Dry-bulb T_{min}/T_{max}, growing season				
T_{min}	-0.218***	0.051	-0.038	-0.042
	(0.0459)	(0.0405)	(0.0603)	(0.0311)
	[0.0459]	[0.0403]	[0.0599]	[0.0311]
T_{max}	-0.225***	-0.046	-0.119*	0.130***
	(0.0562)	(0.0474)	(0.0703)	(0.0341)
	[0.0560]	[0.0471]	[0.0677]	[0.0345]
F-stat ($T_{min} = T_{max}$)	0.01	2.63	0.74	13.34
Observations	1,243	1,211	1,115	1,191
Degree days, growing season				
Degree Days	-0.427***	0.014	-0.108	0.064
	(0.0872)	(0.0662)	(0.0912)	(0.0476)
	[0.0871]	[0.0656]	[0.0895]	[0.0486]
Observations	1,243	1,211	1,115	1,191
Degree days, calendar year				
Degree Days	-0.359***	0.002	-0.102	0.035
	(0.0817)	(0.0566)	(0.0740)	(0.0437)
	[0.0834]	[0.0558]	[0.0737]	[0.0442]
Observations	1,243	1,211	1,115	1,191
Dep. var. mean	5.451	7.142	2.856	5.660

Notes: Each block re-estimates the ICRISAT agricultural decadal-panel specification (Table 1 of the main text) replacing the climate regressors. Outcomes are log levels for the cereal index (rice, wheat, barley, maize, millets, sorghum). Decadal panel (1981–2011) with unit and decade fixed effects and a state × decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.2). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE A15. Census Districts, Long Difference: Individual vs Simultaneous Entry

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
T_{wmin} only				
ΔT_{wmin}	-0.087***	0.092***	-0.022	0.021
	(0.0250)	(0.0201)	(0.0290)	(0.0261)
	[0.0238]	[0.0195]	[0.0279]	[0.0251]
Observations	293	293	293	293
T_{wmax} only				
ΔT_{wmax}	-0.107**	0.109**	-0.047	0.050
	(0.0479)	(0.0428)	(0.0574)	(0.0539)
	[0.0456]	[0.0413]	[0.0551]	[0.0521]
Observations	293	293	293	293
Simultaneous (baseline)				
ΔT_{wmin}	-0.147***	0.159***	-0.009	0.001
	(0.0440)	(0.0329)	(0.0512)	(0.0469)
	[0.0430]	[0.0329]	[0.0497]	[0.0451]
ΔT_{wmax}	0.144*	-0.165**	-0.031	0.049
	(0.0863)	(0.0652)	(0.1004)	(0.0968)
	[0.0848]	[0.0649]	[0.0974]	[0.0936]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	5.42	11.95	0.02	0.12
Observations	293	293	293	293
Simultaneous, no rainfall control				
ΔT_{wmin}	-0.147***	0.155***	-0.009	0.006
	(0.0442)	(0.0333)	(0.0506)	(0.0461)
	[0.0433]	[0.0333]	[0.0492]	[0.0443]
ΔT_{wmax}	0.145*	-0.161**	-0.031	0.044
	(0.0863)	(0.0671)	(0.0996)	(0.0969)
	[0.0849]	[0.0670]	[0.0968]	[0.0938]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	5.44	10.81	0.02	0.08
Observations	293	293	293	293
Dep. var. mean	0.141	-0.049	-0.141	0.049

Notes: Blocks enter T_{wmin} and T_{wmax} one at a time, simultaneously (the baseline decomposition), and simultaneously without the rainfall control. Given the high correlation between the two wet-bulb measures ($\rho = 0.93$), individual entry illustrates the omitted-variable bias the simultaneous specification removes. Outcomes are shares of total district workers. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.2). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.4. Crop composition of the cereal index

The cereal index of Table 1 is a crop-composition average. Table A16 reports the long-difference yield response crop by crop for the three largest grains and the index. Under wet-bulb nighttime warming the index gains (+0.36, significant) while its components pull in opposite directions:

rice and maize gain (imprecisely) and wheat loses (-0.45, at the 10% level), so the aggregate night-yield response is a composition average rather than a uniform crop effect.

TABLE A16. ICRISAT Districts, Long Difference: Log Yields by Crop

	Rice (1)	Wheat (2)	Maize (3)	Cereal Index (4)
Wet-bulb T_{wmin}/T_{wmax} (baseline)				
ΔT_{wmin}	0.269	-0.450*	0.251	0.357**
	(0.2156)	(0.2453)	(0.2794)	(0.1668)
	[0.2038]	[0.2071]	[0.2626]	[0.1601]
ΔT_{wmax}	-0.388	0.833	-0.038	-0.626
	(0.5826)	(0.5357)	(0.5907)	(0.5064)
	[0.5526]	[0.4463]	[0.5537]	[0.4897]
F-stat ($\Delta T_{wmin} = \Delta T_{wmax}$)	0.74	2.79	0.12	2.23
Observations	288	270	272	303
Degree days, growing season				
ΔDegree Days	0.102	0.173	0.331	-0.072
	(0.3731)	(0.2234)	(0.3016)	(0.3021)
	[0.3543]	[0.2007]	[0.2883]	[0.2928]
Observations	288	270	272	303
Dep. var. mean	0.275	0.259	0.387	0.368

Notes: Outcomes are log area-weighted yields by individual crop; the cereal index aggregates rice, wheat, barley, maize, millets, and sorghum weighted by baseline (1966–71) land shares. Long-differenced cross-section: outcomes and climate averaged over 1981–91 (baseline) and 2001–11 (endline), then differenced. All specifications include state fixed effects. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.2). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.5. Heterogeneity by baseline climate and irrigation

Table A17 splits the district panel at the median of baseline nighttime temperature, baseline rainfall, and latitude. The night/day divergence on agricultural wage labour is carried by the cool, dry, and northern halves of the sample ($F = 18.6$, 14.8, and 25.0) and is absent in the hot, wet, and southern halves, where both margins are small and imprecise. Table A18 restricts the ICRISAT panel to districts with above-median predetermined (Green-Revolution era) irrigation shares: the nighttime effect persists within the historically irrigated belt, so nighttime moisture is not a proxy for irrigation.

TABLE A17. Census Districts, Panel: Heterogeneity by Baseline Climate

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
All units				
T_{wmin}	-0.083***	0.086***	-0.013	0.011
	(0.0216)	(0.0173)	(0.0237)	(0.0221)
	[0.0214]	[0.0171]	[0.0235]	[0.0217]
T_{wmax}	0.095***	-0.079***	-0.028	0.009
	(0.0349)	(0.0253)	(0.0348)	(0.0355)
	[0.0352]	[0.0252]	[0.0344]	[0.0351]
F-stat ($T_{wmin} = T_{wmax}$)	10.68	16.27	0.08	0.00
Observations	1,187	1,187	1,187	1,187
Hot (high baseline T_{wmin})				
T_{wmin}	-0.010	0.018	-0.013	-0.001
	(0.0356)	(0.0264)	(0.0317)	(0.0356)
	[0.0348]	[0.0256]	[0.0310]	[0.0348]
T_{wmax}	-0.012	-0.023	-0.028	0.070
	(0.0540)	(0.0391)	(0.0493)	(0.0580)
	[0.0525]	[0.0379]	[0.0481]	[0.0564]
F-stat ($T_{wmin} = T_{wmax}$)	0.00	0.42	0.04	0.63
Observations	600	600	600	600
Cool (low T_{wmin})				
T_{wmin}	-0.120***	0.116***	-0.018	0.026
	(0.0311)	(0.0242)	(0.0370)	(0.0328)
	[0.0307]	[0.0237]	[0.0364]	[0.0321]
T_{wmax}	0.148***	-0.112***	0.003	-0.044
	(0.0457)	(0.0319)	(0.0468)	(0.0408)
	[0.0469]	[0.0317]	[0.0462]	[0.0403]
F-stat ($T_{wmin} = T_{wmax}$)	14.03	18.55	0.07	0.96
Observations	587	587	587	587
Wet (high baseline rain)				
T_{wmin}	-0.053	0.040	0.010	0.005
	(0.0377)	(0.0302)	(0.0401)	(0.0271)
	[0.0365]	[0.0296]	[0.0391]	[0.0267]
T_{wmax}	0.050	-0.019	-0.059	0.025
	(0.0573)	(0.0453)	(0.0606)	(0.0435)
	[0.0553]	[0.0444]	[0.0592]	[0.0429]
F-stat ($T_{wmin} = T_{wmax}$)	1.24	0.65	0.49	0.09
Observations	591	591	591	591
Dry (low rain)				
T_{wmin}	-0.048*	0.089***	-0.062*	0.025
	(0.0288)	(0.0201)	(0.0316)	(0.0330)
	[0.0286]	[0.0199]	[0.0310]	[0.0323]
T_{wmax}	0.060	-0.100***	0.074*	-0.038
	(0.0479)	(0.0316)	(0.0408)	(0.0482)
	[0.0484]	[0.0310]	[0.0395]	[0.0470]
F-stat ($T_{wmin} = T_{wmax}$)	2.16	14.80	3.87	0.65
Observations	596	596	596	596
North (high latitude)				
T_{wmin}	-0.143***	0.146***	-0.014	0.021
	(0.0314)	(0.0252)	(0.0360)	(0.0324)
	[0.0312]	[0.0247]	[0.0355]	[0.0318]
T_{wmax}	0.154***	-0.137***	0.006	-0.035
	(0.0478)	(0.0337)	(0.0477)	(0.0418)
	[0.0490]	[0.0333]	[0.0473]	[0.0412]
F-stat ($T_{wmin} = T_{wmax}$)	15.44	24.96	0.06	0.61
Observations	586	586	586	586
South (low latitude)				
T_{wmin}	-0.072**	0.028	0.023	0.016
	(0.0294)	(0.0304)	(0.0279)	(0.0352)
	[0.0286]	[0.0296]	[0.0273]	[0.0344]
T_{wmax}	0.056	-0.023	-0.070	0.045
	(0.0460)	(0.0424)	(0.0431)	(0.0542)
	[0.0448]	[0.0413]	[0.0422]	[0.0528]
F-stat ($T_{wmin} = T_{wmax}$)	3.16	0.54	1.97	0.11
Observations	601	601	601	601
Dep. var. mean	0.173	0.180	0.323	0.324

Notes: Blocks restrict the estimation sample by baseline (long-run) wet-bulb nighttime temperature, baseline rainfall, and latitude, each split at the median. Outcomes are shares of total district workers. Decadal panel (1981–2011) with unit and decade fixed effects and a state × decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.2). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE A18. ICRISAT Districts, Panel: Baseline Irrigation Subsample

	Cultivated Area (1)	Yield Index (2)	Field Wage (3)	Price Index (4)
All districts				
T_{wmin}	0.090	0.153**	0.137	-0.237***
	(0.0781)	(0.0773)	(0.1243)	(0.0598)
	[0.0778]	[0.0770]	[0.1204]	[0.0598]
T_{wmax}	-0.579***	-0.194	-0.328*	0.366***
	(0.1534)	(0.1280)	(0.1928)	(0.0936)
	[0.1529]	[0.1270]	[0.1854]	[0.0946]
F-stat ($T_{wmin} = T_{wmax}$)	9.01	3.07	2.28	16.59
Observations	1,243	1,211	1,115	1,191
Above-median baseline irrigation share				
T_{wmin}	0.126	0.240*	0.382**	-0.177**
	(0.0941)	(0.1269)	(0.1830)	(0.0810)
	[0.0929]	[0.1249]	[0.1747]	[0.0797]
T_{wmax}	-0.725***	-0.281	-0.762***	0.348**
	(0.1720)	(0.1810)	(0.2799)	(0.1428)
	[0.1709]	[0.1785]	[0.2665]	[0.1419]
F-stat ($T_{wmin} = T_{wmax}$)	11.41	3.11	6.55	5.82
Observations	664	636	586	633
Dep. var. mean	5.451	7.142	2.856	5.660

Notes: The second block restricts the sample to districts with above-median predetermined (Green-Revolution era) irrigation shares. The persistence of the T_{wmin} yield dividend within the irrigated belt indicates nighttime moisture is not a proxy for irrigation. Decadal panel (1981–2011) with unit and decade fixed effects and a state $\times$ decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.2). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.6. Town sample stability

The town long difference of Table A9 uses the balanced set of towns present in every round since 1981. Table A19 reports the full (unbalanced) decadal town panel, with and without a predetermined town-size control (1961 population interacted with a linear decade trend) that allows initially larger towns their own transformation trajectory. The nighttime pull onto town agricultural wage labour persists in the unbalanced panel (+0.03) with the seasonal share moving against it, while the non-agricultural coefficients are small and imprecise on both margins; the town-size control leaves every coefficient unchanged.

TABLE A19. Census Towns, Panel: Baseline-Size-Specific Trends

	Seasonal (1)	Ag. Labour (2)	Cultivation (3)	Non-Agriculture (4)
Baseline				
T_{wmin}	-0.023**	0.030**	-0.017*	0.008
	(0.0113)	(0.0118)	(0.0094)	(0.0160)
	[0.0106]	[0.0100]	[0.0084]	[0.0148]
T_{wmax}	0.046**	-0.025	0.016	-0.037
	(0.0184)	(0.0190)	(0.0133)	(0.0258)
	[0.0177]	[0.0165]	[0.0125]	[0.0237]
F-stat ($T_{wmin} = T_{wmax}$)	5.91	3.37	2.15	1.29
Observations	16,215	16,215	16,215	16,215
Adding 1961 population × decade trend				
T_{wmin}	-0.023**	0.031***	-0.016*	0.008
	(0.0113)	(0.0118)	(0.0093)	(0.0159)
	[0.0106]	[0.0100]	[0.0084]	[0.0147]
T_{wmax}	0.047**	-0.026	0.015	-0.036
	(0.0184)	(0.0189)	(0.0133)	(0.0258)
	[0.0177]	[0.0165]	[0.0125]	[0.0238]
F-stat ($T_{wmin} = T_{wmax}$)	5.99	3.50	1.98	1.19
Observations	16,215	16,215	16,215	16,215
Dep. var. mean	0.095	0.094	0.077	0.732

Notes: The second block adds a predetermined town-size control: 1961 population interacted with a linear decade trend, allowing initially larger towns their own transformation trajectory. Sample is the full set of towns observed from 1981 onward. Decadal panel (1981–2011) with unit and decade fixed effects and a state × decade linear trend. Temperatures are wet-bulb unless noted, constructed as a moving average of the preceding decade's growing seasons (June–February), as inverse-distance-squared weighted averages of surrounding IMD grid points (see Appendix C.2). All specifications control for total precipitation over the same window unless noted. Standard errors in parentheses, clustered at the district level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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